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(19) **United States**(12) **Patent Application Publication**
ALVES et al.(10) **Pub. No.: US 2025/0282832 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **CHIMERIC NUCLEOTIDE SEQUENCE,
VECTOR FOR EXPRESSION IN MAMMALS,
RNA VACCINE, CHIMERIC FUSION
PROTEIN, USE IN THE PRODUCTION OF A
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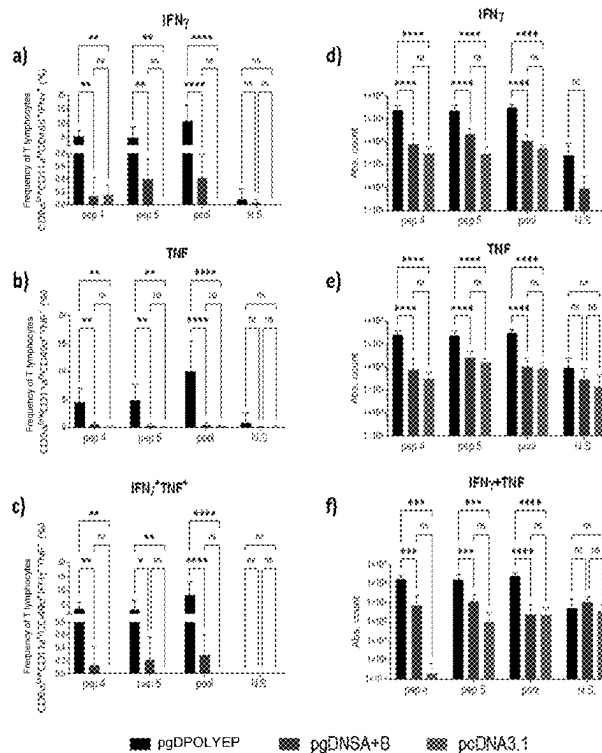
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(57)

ABSTRACT

A chimeric nucleotide sequence that corresponds to an encoded fusion protein comprising a polypeptide resulting from selecting and juxtaposing multiple epitopes from a coronavirus protein to induce an immune response in mammals. In one embodiment, said fusion protein comprises: a) a first peptide consisting of epitopes found in the amino acid sequence of replicase polyprotein 1ab (PR1ab); b) a first spacer; c) a modified form of herpes simplex virus type 1 (HSV-1) glycoprotein D (gD). In one embodiment, the replicase polyprotein is defined by SEQ ID NO: 96 flanked by a gD fragment comprising the amino acid sequence defined by SEQ ID NO: 98 in the N-terminal portion and another gD fragment comprising the amino acid sequence defined by SEQ ID NO: 100 in the C-terminal region. Use of the fusion protein has surprising results in inducing cellular and humoral immune responses against coronavirus, SARS-COV-2, and related viruses.

Specification includes a Sequence Listing.

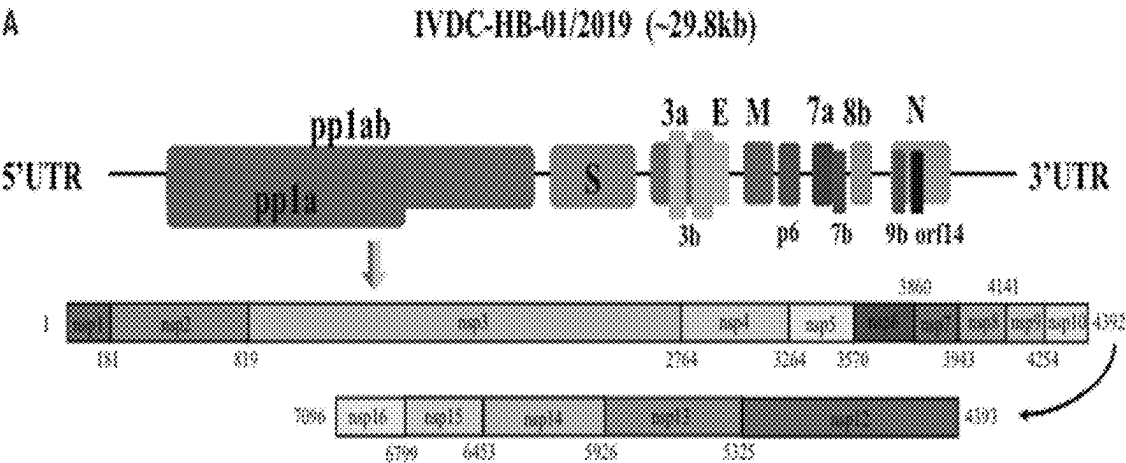


FIG. 1

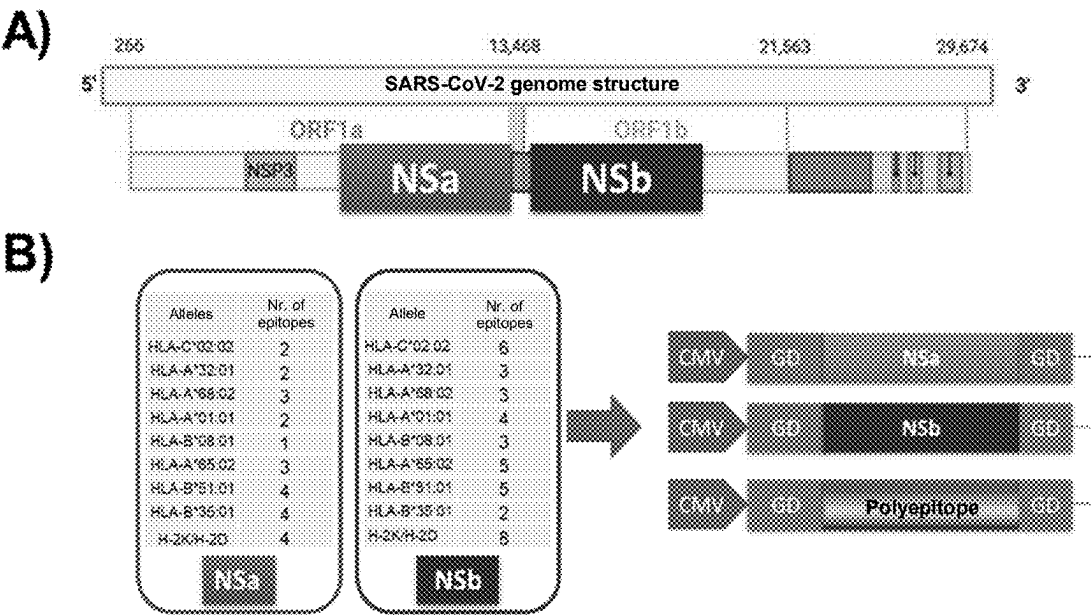


FIG. 2

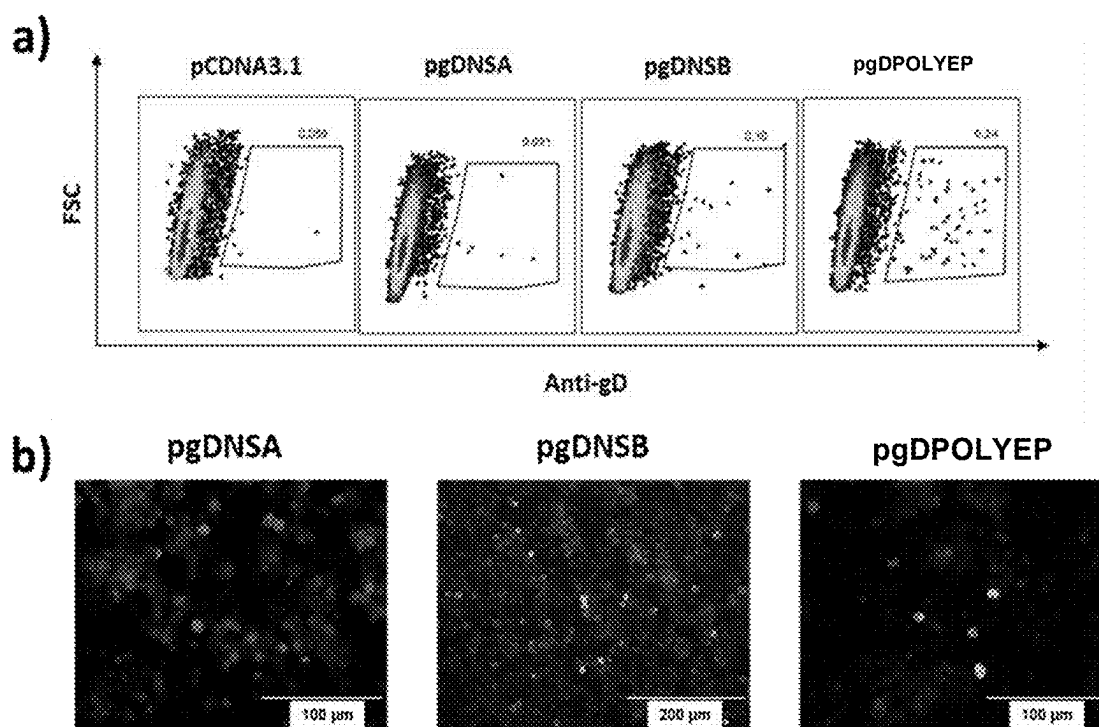


FIG. 3

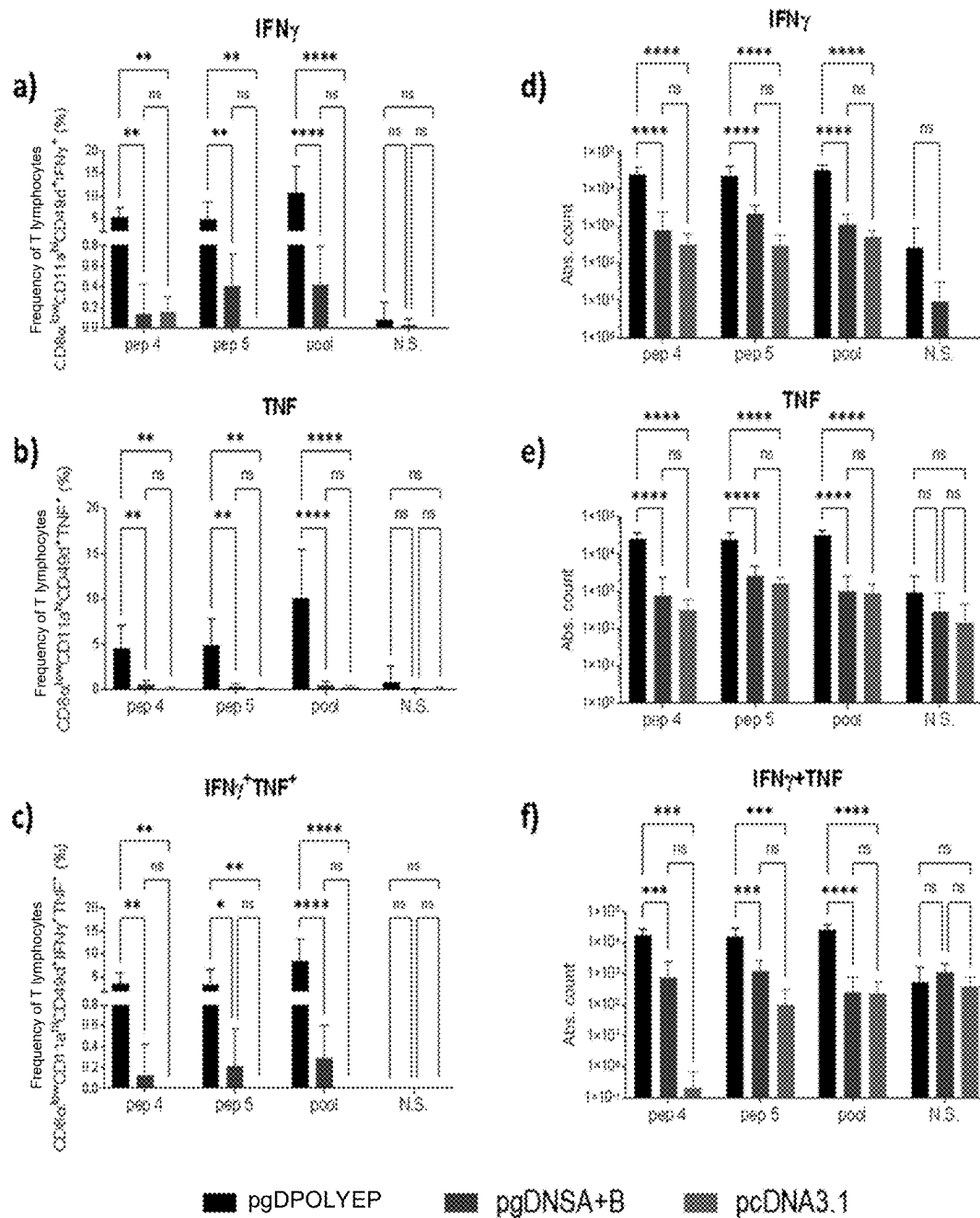


FIG. 4

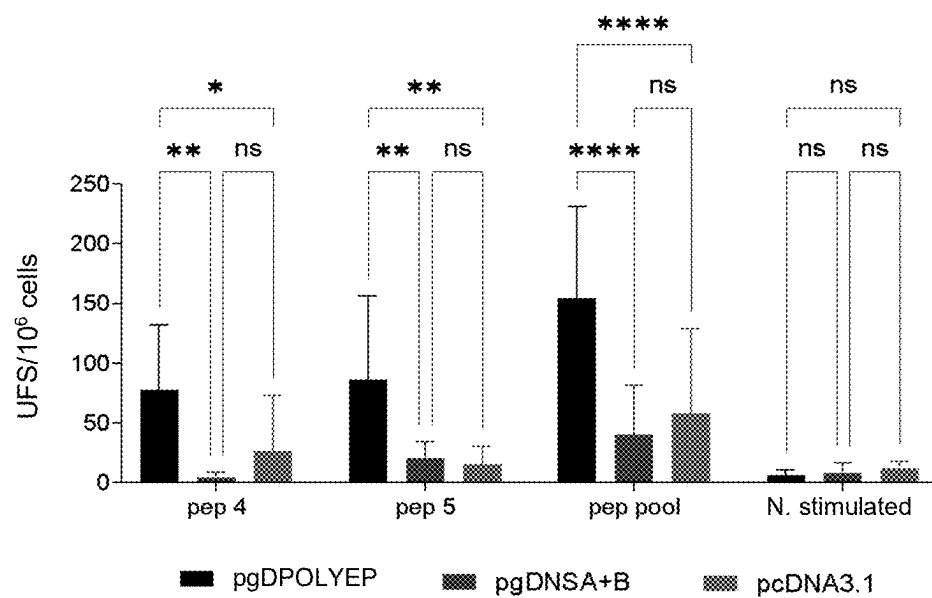


FIG. 5

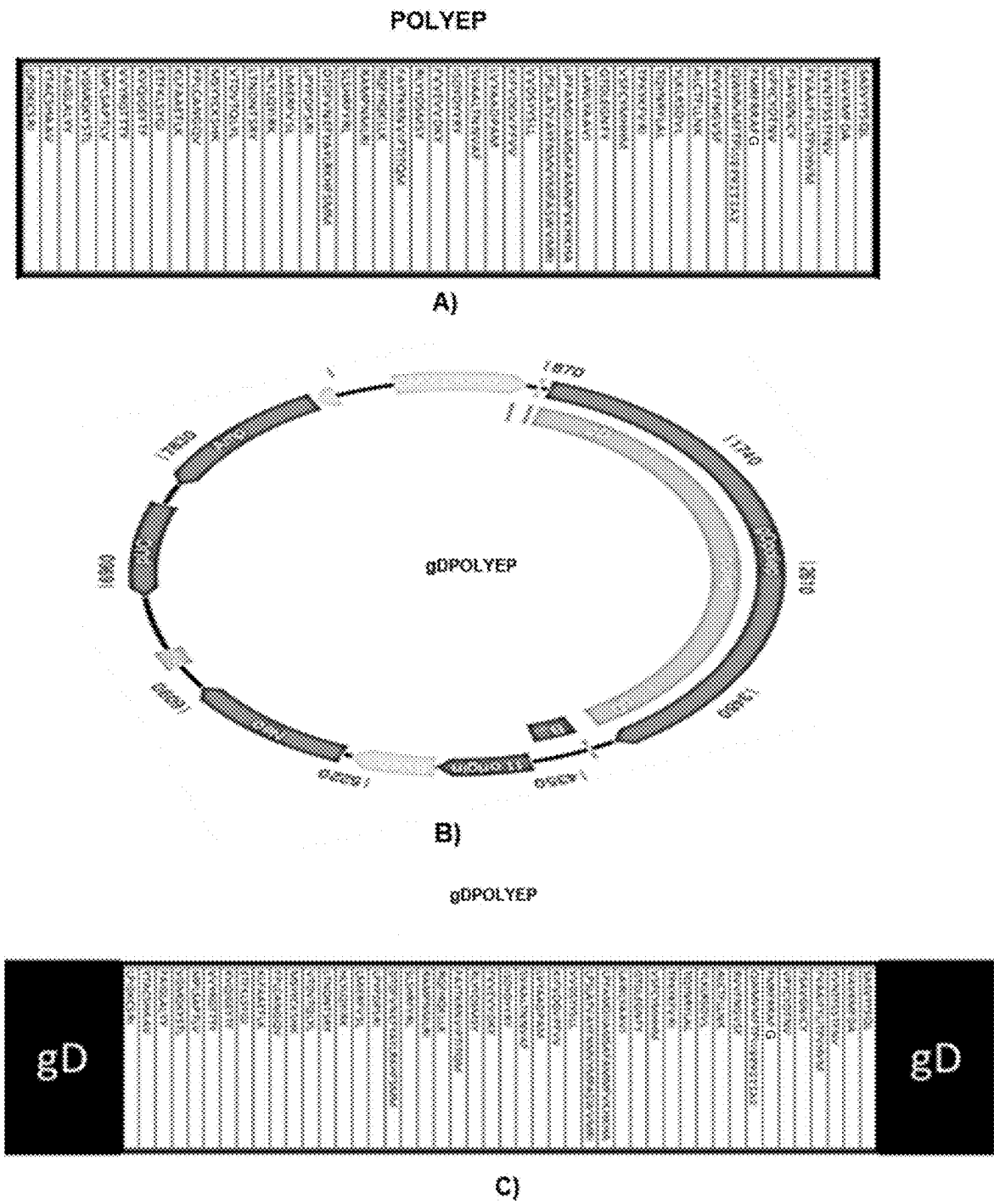


FIG. 6

**CHIMERIC NUCLEOTIDE SEQUENCE,
VECTOR FOR EXPRESSION IN MAMMALS,
RNA VACCINE, CHIMERIC FUSION
PROTEIN, USE IN THE PRODUCTION OF A
VACCINE AGAINST CORONAVIRUS**

SEQUENCE LISTING

[0001] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Apr. 24, 2024, is named 61094_001U1_SL.txt and is 107,948 bytes in size.

FIELD OF THE INVENTION

[0002] The present invention belongs to the technical field of Biotechnology. More specifically, a chimeric nucleotide sequence is described, the corresponding encoded fusion protein, which comprises a polypeptide resulting from the selection and juxtaposition of multiple epitopes of a coronavirus protein to induce immune response in mammals. In one embodiment, said fusion protein comprises: a) a first peptide consisting of epitopes present in the amino acid sequence of replicase polyprotein 1ab (PR1ab); b) a first spacer; c) a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1). In one embodiment, the replicase polyprotein is defined by SEQ ID NO: 96 flanked from a gD fragment comprising the amino acid sequence defined by SEQ ID NO: 98 in the N-terminal portion and another gD fragment comprising the amino acid sequence defined by SEQ ID NO: 100 in the C-terminal region. The use of the fusion protein shows surprising results for the induction of cellular and humoral immune response against coronavirus, SARS-COV-2, and related viruses.

BACKGROUND OF THE INVENTION

[0003] The re-emergence of SARS-COV-2 betacoronavirus has so far resulted in more than 107 million infections and more than two million deaths worldwide (Amanat & Krammer, 2020). The rapid spread of this virus associated with a high morbidity rate, as well as the lack of scientifically confirmed therapeutic and/or prophylactic strategies reinforce the urgency of developing vaccines. The pandemic nature of COVID-19 has driven the race to develop therapeutic treatments, so that dozens of drugs and vaccine candidates are being tested in the clinical and preclinical context (W.-H. Chen et al., 2020; Sun et al., 2020).

[0004] Symptoms of COVID-19 can range from absent and mild to severe, and includes fever, dry cough, dyspnea, myalgia, tiredness, leukopenia, and even pneumonia. The patient immune system widely contributes to this condition. In more severe cases, a phenomenon called cytokine storm, an abrupt and intense release of pro-inflammatory cytokines (IFN- α , IFN- γ , IL-1 β , IL-6, IL-12, IL-18, IL-33, TNF- α , TGF- β) and chemokines (CCL2, CCL3, CCL5, CXCL8, CXCL9, CXCL10) result in an accumulation of fluid in the lungs, multiple failure and death (Li et al., 2020). The virus causing the disease belongs to the same family as MERS-COV and SARS-COV, being responsible for acute respiratory syndromes that have become epidemics in the last two decades.

[0005] SARS-COV-2 belongs to the Coronaviridae family, order Nidovirales, which is divided into four genera (Alpha-, Beta-, Gamma-, Delta-Coronavirus), most of which are

zoonotic viruses (Grifoni et al., 2020). These viruses are enveloped with positive sense single-stranded RNA, and genomes between 26 and 32 kb with 29.8 kb, the SARS-COV-2 genome has 14 ORFs (Open Reading Frame(s)) that encode 27 proteins. Region 5' contains the overlapping ORFs orf1ab and orf1a that encode two polypeptides that are processed into 15 non-structural proteins (nsp1-nsp10 and nsp12-nsp16), and region 3' contains the 4 structural proteins (spike, envelope, membrane, and nucleocapsid), and 8 accessory proteins (3a, 3b, p6, 7a, 7b, 8b, 9b, and orf14) (Wu et al., 2020), as shown in FIG. 1.

[0006] Electron microscopy tests showed that SARS-COV and SARS-COV-2 show great similarity and enter the cells using the same receptor, the angiotensin converting enzyme (ACE2), to which the spike glycosylated protein binds and undergoes irreversible conformational changes before entering the cell (Wrapp et al., 2020). S glycoprotein of SARS-COV-2 has 77% similarity to SARS-COV spike and it is a natural target due to its critical role in receptor binding and consequent fusion of the virus membranes and the host, being therefore considered the main antigen (Yuan et al., 2020).

[0007] Most vaccine strategies under development target the viral spike protein, which is present on the virion surface and is responsible for the virus interaction with the ACE2 cellular receptor, being, therefore, the main target of neutralizing antibodies (Roper & Rehm, 2009). However, in the natural response developed against coronavirus, there is not always an induction of antigen-specific long-lasting antibodies. Preclinical studies with vaccine strategies based on inactivated SARS-COV-1 showed that, even with the induction of antibodies with neutralizing and protective capacity, some of the immunized animals showed clinical complications after the challenge, with pulmonary injury with eosinophilic infiltration being reported in mice, and increased severity of the disease being reported in ferrets (Tseng et al., 2012; Weingartl et al., 2004). Epidemiological data from the SARS-COV-1 outbreak in China in 2005 suggests that cross-reaction neutralizing antibodies, detectable during the first two weeks of the disease, were correlated with high rates of early mortality by SARS (Yang et al., 2005). Further studies have shown that antibodies generated against S protein variants tend to increase infection of human macrophages by a mechanism known as Antibody-Dependent Enhancement (ADE) (Wang et al., 2014). This phenomenon is the pathogenic basis of the feline infectious peritonitis virus (FIPV, a type II coronavirus) and severe dengue (Tirado & Yoon, 2003).

[0008] Based on data from the dengue literature, it is known that when the vaccine response is accompanied by the activation of CD8⁺ T cells, the pathogenic effect of the antibodies is neutralized (Zellweger et al., 2014). In addition, according to previous studies using SARS-COV-1, the adoptive transfer of CD8⁺ T cells promotes viral load reduction, clinical recovery, and protection under experimental conditions (Zhao et al., 2010). In convalescent patients, it was possible to detect CD8⁺ and CD4⁺ memory T cells up to four years after primary infection, which were responsive to in vitro restimulation, and resulted in the high production of cytokines and molecules having an antiviral role, such as IFN γ , perforin and granzyme B (H. Chen et al., 2005). These results show that T cells play a critical role in the mediation of protection against SARS-COV-2.

[0009] Cytotoxic T lymphocytes (CTL) recognize antigens as short peptides (usually about 8 to about 10 amino acids in length) presented by class I molecules of the main histocompatibility complex (MHC) (Townsend & Bodmer, 1989). The characterization of these antigen fragments, called immunogenic epitope determinants, has allowed the design of immunization strategies based on the use of vaccines containing such segments (Berzofsky et al., 2001; Melief & Van Der Burg, 2008). As a product of recombinant DNA technology, it was possible to develop a class of new biomolecules with chimeric characteristics and multi-antigen property. By genetically fusing two or more antigenic segments of one or more target proteins, the product obtained may have the immunogenic determinants of the different original target proteins. Thus, the fusion of different epitopes from various sources can induce specific multivalent immunological response against different pathogens/antigens through a unique construction, such strategy being called polypeptide or multi-epitope vaccine. In general, vaccine strategies that employ this method have been studied for the treatment of infectious diseases and cancer. There are some works that use this strategy to create COVID-19 vaccines. However, the strategies developed have structural proteins as a target (Kar et al., 2020; Naz et al., 2020; Oladipo et al., 2021; Safavi et al., 2020; Saha et al., 2021; Sohail et al., 2021; Tahir Ul Qamar et al., 2020).

[0010] Glycoprotein D (gD) has the ability to interact with the HVEM receptor (Herpesvirus Entry Mediator), which promotes the activation of immune system cells directly by the production of NF- κ B (Sciortino et al., 2008), or indirectly by blocking BTLA (B and T Lymphocyte Attenuator) and CD160 binding to this receptor that would result in co-inhibitory signals (Steinberg et al., 2011). It is known in the art the use of antigens fusion to gD for the development of therapeutic vaccines against tumors induced by the human papillomavirus (HPV) both in the form of DNA vaccine (Diniz et al., 2010; Diniz et al., 2013; APS et al., 2015; Diniz et al., 2016) and in the form of purified protein vaccine (Porchia et al., 2011; Porchia et al., 2017), whose results show a significant increase in the induction of local and systemic CD8⁺ T lymphocytes specific against tumor antigens genetically fused to gD.

[0011] When searching the state of the art in scientific and patent literatures, the following documents were found that relate to the subject:

[0012] PI 0904880-4 discloses the construction of two plasmids, one containing gene expressing oncoprotein E7 of the human papillomavirus type-16 (HPV-16) fused to the glycoprotein D (gD) of herpes simplex type 1 (HSV-1) and another plasmid containing the gene expressing interleukin-2, intended to control tumors.

[0013] PI 1003749-7 discloses a hybrid protein, formed by the genetic fusion of oncoprotein E7 of the human papillomavirus type-16 (HPV-16) with a modified form of the glycoprotein D of herpes simplex type 1, intended to be a vaccine adjuvant to other antigens and/or as an active ingredient in pharmaceutical compositions to control tumors.

[0014] U.S. Pat. Nos. 8,962,816 and 9,724,406 disclose chimeric proteins, in which one or more antigens are introduced in the C-terminal region of a glycoprotein D (gD) of herpes simplex type 1, intended to more strongly enhance an individual's immune response to the antigen when compared to the protein without fusion with gD. These documents

disclose possible applications for vaccines against influenza, malaria, cervical cancer, and HIV/AIDS.

[0015] The scientific literature that circumvents the invention, without however anticipating or suggesting it, includes the documents below.

[0016] Amanat, F., & Krammer, F. (2020). Perspective SARS-COV-2 Vaccines: Status Report. <https://doi.org/10.1016/j.immuni.2020.03.007>.

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- [0052] From what is evident from the literature researched, no documents were found anticipating or suggesting the teachings of the present invention, so that the solution proposed herein has, in the eyes of the inventors, industrial applicability, novelty and inventive activity relative to the state of the art. One of the objects of the present invention provides an alternative vaccine and/or an alternative to obtain vaccines against Coronavirus, SARS-COV-2, or COVID-19, and related viruses.
- ### SUMMARY OF THE INVENTION
- [0053] The present invention provides a chimeric nucleotide sequence comprising a polypeptide resulting from the selection and juxtaposition of multiple epitopes of a coronavirus protein to induce immune response in mammals.
- [0054] The DNA or RNA nucleotide sequences encoding the aforementioned polypeptide are also the object of the invention, as well as the mammalian expression vectors comprising said nucleotide sequences.
- [0055] The present invention provides an alternative vaccine and/or an alternative to obtain cell and/or humoral immunity-inducing vaccines against Coronavirus, SARS-COV-2, or COVID-19, and related viruses. In one embodiment, the antigenic elements are non-structural proteins of the virus.
- [0056] In one embodiment, the present invention provides a vaccine to induce a combined immunological response of CD8+ T cells producing antiviral molecules in addition to the humoral response against SARS-COV-2.
- [0057] The in vivo use of the present invention provides a surprising potential to promote viral load reduction, clinical recovery and protection against coronavirus, SARS-COV-2, or COVID-19, and related viruses, among other advantages.
- [0058] In one embodiment, the antigenic elements, or targets of the vaccine of the invention are found in the non-structural proteins encoded by the ORF1ab of the Brazilian strain of SARS-COV-2, in contrast to the technologies existing in the art that focus mainly on the virus Spike protein (S protein).
- [0059] In one embodiment, the present invention provides the association of immunogenic epitopes of these regions compared with the HLAs (human leukocyte antigen system) most frequently found in the Brazilian population, being: C*02:02, A*32:01, A*68:02, A*01:01, B*08:01, A*65:02, B*51:01, B*35:01; where each allele is frequent in at least 3% of the population of Brazil. In other embodiments, the association of immunogenic epitopes is compared with HLAs of specific ethnic groups and customized to different populations.
- [0060] In one embodiment, separators are used, preferably defined by the amino acid sequence defined by SEQ ID NO: 94 added between the peptides. Among other reasons, these separators may facilitate the structuring of the antigen after the intracellular translation into the expression and/or to facilitate antigen processing.
- [0061] In one embodiment, the sequence containing the multiepitope is inserted in the sequence of the glycoprotein D (gD) of the herpesvirus. Among other advantages, gD has an adjuvant effect for the activation of T lymphocytes, related to its ability to interact with HVEM (Herpesvirus Entry Mediator) receptor that promotes activation of the immune system cells.
- [0062] The present invention presents the following objects:
- [0063] In a first object, the present invention provides a chimeric nucleotide sequence comprising the sequences encoding, in phase, a plurality of epitopes of a coronavirus protein.
- [0064] In one embodiment, said coronavirus protein is a non-structural protein of the virus, such as the replicase protein 1ab. In one embodiment, the sequences encoding the replicase 1ab multiepitope are aligned in phase, so that the encoded product is a chimeric protein of a replicase 1ab polypeptide.
- [0065] In one embodiment, said nucleotide sequence additionally comprises a nucleotide sequence that encodes a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).
- [0066] In one embodiment, said nucleotide sequence additionally comprises a nucleotide sequence encoding a first spacer polypeptide.
- [0067] In one embodiment, said nucleotide sequence comprises:
- [0068] a) a nucleotide sequence that encodes a first peptide comprising epitopes found in the amino acid sequence of the replicase polypeptide 1ab (PR1ab);
 - [0069] b) a nucleotide sequence encoding a first spacer polypeptide; and
 - [0070] c) a nucleotide sequence that encodes a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).
- [0071] In one embodiment, the nucleotide sequence encoding the polypeptide comprises a phased combination of the nucleotide sequences selected from SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 73, 75, 77, 79, 81, 83, 85, 87, 89, 91.
- [0072] In one embodiment the nucleotide sequence encoding the polypeptide is defined by SEQ ID NO: 95.

[0073] In one embodiment, the nucleotide sequence encoding the modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1) is defined by SEQ ID NO: 97 and/or SEQ ID NO: 99.

[0074] In a second object, an expression vector is provided, which comprises:

[0075] the aforementioned chimeric nucleotide sequence encoding the polypeptide; and

[0076] one or more expression promoters functionally linked to said nucleotide sequence.

[0077] In one embodiment, the polypeptide sequence of the polypeptide comprises a combination of the polypeptide sequences selected from SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92.

[0078] In a third object, an RNA vaccine is provided, which comprises an RNA sequence that encodes the aforementioned polypeptide.

[0079] In one embodiment, said RNA sequence comprises a combination of the RNA sequences selected from SEQ ID NO: 113 to SEQ ID NO: 158. In one embodiment, said RNA sequence additionally comprises SEQ ID NO: 105. In one embodiment, said RNA sequence additionally comprises SEQ ID NO: 161 and/or SEQ ID NO: 162.

[0080] In a fourth object, the present invention provides a chimeric polypeptide sequence comprising a plurality of epitopes of a coronavirus protein.

[0081] In one embodiment, said coronavirus protein is a non-structural protein of the virus, such as the replicase protein 1ab. In one embodiment, multiple epitopes of replicase 1ab are aligned, thus forming a chimeric protein of a replicase 1ab polypeptide.

[0082] In one embodiment, said polypeptide sequence additionally comprises a polypeptide sequence of a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

[0083] In one embodiment, said polypeptide sequence additionally comprises a sequence of a first spacer polypeptide.

[0084] In one embodiment, the chimeric fusion protein comprises:

[0085] a) a first peptide consisting of epitopes found in the amino acid sequence of the replicase polypeptide 1ab (PR1ab);

[0086] b) a first spacer; and

[0087] c) modified form of glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

[0088] In one embodiment, the peptide sequence comprising the polypeptide is defined by SEQ ID NO: 96.

[0089] In one embodiment, the peptide sequence of the modified form of gD is defined by SEQ ID NO: 98 and/or SEQ ID NO: 100.

[0090] As a fifth object, it is provided the use of the chimeric nucleotide sequence, the RNA chimeric sequence or the fusion protein for the preparation of a vaccine for coronavirus, SARS-CoV-2, and/or related viruses.

[0091] These and other objects of the invention will be immediately valued by those skilled in the art and will be described in detail below.

BRIEF DESCRIPTION OF THE FIGURES

[0092] The following figures are shown:

[0093] FIG. 1 depicts the organization of the SARS-CoV-2 genome showing the arrangement of the ORFs (Open Reading Frame) and the distribution of structural and non-structural proteins (Wu et al., 2020).

[0094] FIG. 2 shows a) a scheme of the gene structure of SARS-CoV-2 and b) the target alleles of two regions: NSa and NSb.

[0095] FIG. 3 shows the analysis of the in vitro expression of vaccine antigens. After 36 hours of transfection with each vaccine plasmid, the cells were subjected to gD protein immunodetection. In (a), a flow cytometry analysis was performed. In (b), after fixing the cell monolayer on the plate, immunofluorescence was performed with glycoprotein D antibody, where cells positive for the expression of the recombinant antigen encoded by DNA vaccines are marked in green and the nucleus of the cells are marked in blue, with an increase of 20× or 40×.

[0096] FIG. 4 shows the evaluation of cellular response induced by DNA vaccines associated with ICS electroporation. 5-6 week-old C57BL/6 mice received two intramuscular doses of the vaccines (50 ug pgDPOLYEP, 50 ug NSA+50 ug NSB or 50 ug pcDNA3.1) together with electroporation, with two weeks between the applications. Two weeks after the last dose, the animals were euthanized and the spleen cells were used for ex vivo stimulation with SARS-CoV-2 peptides for functional evaluation (a). The subpopulation of CD8⁺ T lymphocytes (CD3⁺CD8⁺) that experienced the encounter with the in vivo antigen (CD49d⁺CD11a^{hi}) was evaluated for the relative frequency of IFN γ (a)-, TNF (b)-, and IFN γ +TNF (c)-producing cells, as well as the absolute count of the IFN γ (d)-, TNF (e)-, and IFN γ +TNF (f)-producing cells. Data represents the mean $\pm$ SEM and one-tailed Mann-Whitney U test, where *p<0.05 **p<0.01, ***p<0.001, ****p<0.0001.

[0097] FIG. 5 shows the evaluation of cellular response induced by DNA vaccines associated with ELISPOT electroporation. 5-6 week-old C57BL/6 mice received two intramuscular doses of the vaccines (50 ug pgDPOLYEP, 50 ug NSA+50 ug NSB or 50 ug pcDNA3.1) together with electroporation, with two weeks between the applications. Two weeks after the last dose, the animals were euthanized and the spleen cells were used for ex vivo stimulation with SARS-CoV-2 peptides for evaluating IFN- γ production. Data represents the mean $\pm$ SEM and the one-tailed Mann-Whitney U test, where *p<0.05 **p<0.01, ****p<0.0001.

[0098] FIG. 6 shows a schematic representation of an embodiment of the invention, in which: in A) an embodiment of the chimeric polypeptide sequence of the invention (POLYEP) is schematically shown, which comprises the polypeptide sequences of a polypeptide of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92; in B) the chimeric nucleotide sequence of the invention is shown in the embodiment which comprises SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 73, 75, 77, 79, 81, 83, 85, 87, 89, 91 and two nucleotide sequences of gD fragments, one in 5' and the other in 3', these sequences being functionally linked to a plasmid expression vector (pgDPOLYEP) functioning as a DNA vaccine and comprising a CMV promoter; in C) an embodiment of the chimeric polypeptide sequence of the invention (gDPOLYEP) is schematically shown, which comprises the

polypeptide sequences of a polyepitope of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92 flanked from a gD fragment comprising the amino acid sequence defined by SEQ ID NO: 98 in the N-terminal portion and another gD fragment comprising the amino acid sequence defined by SEQ ID NO: 100 in the C-terminal region.

DETAILED DESCRIPTION OF THE INVENTION

[0099] The present invention provides an alternative vaccine and/or an alternative to obtain cell and/or humoral immunity-inducing vaccines against Coronavirus, SARS-COV-2, or COVID-19, and related viruses.

[0100] The present invention provides a chimeric nucleotide sequence comprising the sequences encoding, in phase, a plurality of epitopes of a coronavirus protein, and/or of the fusion chimeric polypeptide.

[0101] In one embodiment, said coronavirus protein is a non-structural protein of the virus, such as the replicase protein 1ab. In one embodiment, the sequences encoding multiple epitopes of replicase 1ab are aligned in phase. The encoded product is a chimeric protein from a replicase 1ab polypeptide.

[0102] In one embodiment, the nucleotide sequence encoding the polyepitope comprises a phased combination of the nucleotide sequences selected from SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 73, 75, 77, 79, 81, 83, 85, 87, 89, 91.

[0103] In one embodiment, the nucleotide sequence encoding the polyepitope is defined by SEQ ID NO: 95.

[0104] In one embodiment, said nucleotide sequence additionally comprises a nucleotide sequence that encodes a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

[0105] In one embodiment, the nucleotide sequence encoding the modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1) is defined by SEQ ID NO: 97 and/or SEQ ID NO: 99.

[0106] In one embodiment, said nucleotide sequence additionally comprises a nucleotide sequence encoding a first spacer polypeptide.

[0107] In one embodiment, said chimeric nucleotide sequence comprises:

[0108] a) a nucleotide sequence that encodes a first peptide comprising epitopes found in the amino acid sequence of the replicase polyprotein 1ab (PR1ab);

[0109] b) a nucleotide sequence encoding a first spacer polypeptide; and

[0110] c) a nucleotide sequence that encodes a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

[0111] In one embodiment, said chimeric nucleotide sequence is defined by SEQ ID NO: 101.

[0112] A mammalian expression vector, comprising said chimeric nucleotide sequence, is also part of the objects of the invention.

[0113] In one embodiment, said expression vector is a DNA vaccine to induce cellular and humoral immunity, having as antigenic targets the non-structural proteins of the virus, inducing a combined immunological response of

CD8⁺ T cells producing antiviral molecules in addition to the humoral response against SARS-COV-2.

[0114] In one embodiment, the use of the invention surprisingly promotes viral load reduction, clinical recovery and protection against coronavirus, SARS-COV-2, or COVID-19, and related viruses.

[0115] In one embodiment, said SARS-COV-2 antigens were genetically fused to another viral protein, the glycoprotein D (gD) of the Herpes Simplex Virus Type 1 (HSV-1). This fusion showed an increase in antigens immunogenicity. gD showed adjuvant effect for the activation of T lymphocytes.

[0116] The present invention provides an RNA vaccine comprising an RNA sequence that encodes a plurality of epitopes of a coronavirus non-structural protein.

[0117] In one embodiment, said vaccine comprises the sequences encoding multiple epitopes of replicase 1ab aligned in phase.

[0118] In one embodiment, said vaccine comprises a combination of the RNA sequences selected from SEQ ID NO: 113 to SEQ ID NO: 158.

[0119] The present invention provides a chimeric fusion protein comprising a plurality of epitopes of a coronavirus non-structural protein.

[0120] The present invention provides a chimeric fusion protein comprising a combination of the peptide sequences of replicase 1ab selected from SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92.

[0121] In one embodiment, said chimeric fusion protein comprises SEQ ID NO: 96.

[0122] In one embodiment, said chimeric fusion protein additionally comprises a polypeptide sequence of a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

[0123] In one embodiment, said polypeptide sequence of the modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1) is defined by SEQ ID NO: 98 and/or SEQ ID NO: 100.

[0124] In one embodiment, said chimeric fusion protein additionally comprises a sequence of a first spacer polypeptide, preferably defined by SEQ ID NO: 94.

[0125] In one embodiment, said fusion protein comprises:

[0126] a) a first peptide consisting of epitopes found in the amino acid sequence of the replicase polyprotein 1ab (PR1ab);

[0127] b) a first spacer;

[0128] c) modified form of glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

[0129] In one embodiment, the first peptide consists in epitopes present in the amino acid sequence of the NSa regions as defined in SEQ ID NO: 110 and/or NSb as defined in SEQ ID NO: 112 in PR1ab.

[0130] In one embodiment, the spacer consists in the amino acid sequence GGGS as defined in SEQ ID NO: 94.

[0131] In one embodiment, the invention provides a vector comprising the sequences encoding the fusion protein defined above. In one embodiment, said vector additionally comprises a promoter region and Kozak sequence, in which the promoter is preferably CMV and said vector is pcDNA3.1.

[0132] The results shown in this patent application demonstrate that, surprisingly, mice that received the vaccine of

the invention, in the embodiment of pgDPOLYEP DNA, exhibited CD8⁺ T cells with a dominant Th1 phenotype once stimulated with peptides (pep), with significantly higher numbers of IFN γ -, TNF- and IL-2-producing cells, compared to the other DNA vaccines containing the NSA and NSB portion or the pcDNA3.1 vector.

[0133] The vaccine of the invention, in the embodiment comprising the chimeric nucleotide sequences encoding the fusion protein (called pgDPOLYEP), induced the highest production of IFN γ when compared to the formulations where pgDNA+pgDNSB or the empty vaccine vector pDNA3.1 was administered. Altogether, the results show that pgDPOLYEP is immunogenic capable of inducing cell-type immune response by inducing IFN γ - and TNF-producing CD8⁺ T lymphocytes.

EXAMPLES

[0134] The examples shown here are intended only to exemplify one of the numerous ways to realize the invention, but without limiting the scope of the invention.

Example 1—Construction of the pgDPOLYEP Vaccine

[0135] For the choice of target sequences within the non-structural proteins of Covid19, an associative analysis was performed between the predicted epitopes and the most frequent HLA type I molecules found in the Brazilian population. For the same chosen regions, the linking epitopes of MHC H-2Kb (C57BL/6) were predicted. The

allele frequency of HLA class I found in the Brazilian population recovered was from the NCBI database (<https://www.ncbi.nlm.nih.gov/projects/gv/mhc/ihwg.cgi>). Frequent alleles in at least 3% of the Brazilian population were selected (HLA: C*02:02, A*32:01, A*68:02, A*01:01, B*08:01, A*65:02, B*51:01, B*35:01) (Dos Santos Franco et al., 2017). The epitopes within the amino acid sequence of the replicase 1ab (PR1ab) polyprotein were predicted using the IEDB (Immune epitope database) analysis resource (<http://tools.immuneepitope.org/mhci/>), according to the recommended method. The epitopes predicted within the PR1ab were classified by their percentage value and for each HLA molecule considered, and the best 5 ones were selected with a percentile less than 0.5. Epitopes with 8, 9, 10, and 11 amino acids in length for the HLA class I were considered.

[0136] After the associative analysis, two regions in PR1ab were selected that concentrate most of the target sequences for humans and mice (NSa and NSb). The nucleotide sequence encoding both regions, as well as that of the peptides predicted in sequence and linked by the sequence encoding the GGG amino acids (SEQ ID NO: 94), were inserted in the vaccine vector pcDNA3.1 in fusion with the glycoprotein D (gD) of the herpesvirus, with the addition of Kozak sequence and under the CMV promoter control. The three plasmids (pCovid_NSa, pCovid_NSb, and pCovid_Polyep), respectively comprising Seq ID 1, 2, and 3, as well as the corresponding encoded peptides (purified by reverse phase HPLC with >95% purity), were synthesized and commercially obtained (Genscript), FIG. 2, and Table 1.

TABLE 1

Peptides selected in PR1ab of SARS-CoV-2 (GeneBank: MT126808.1)						
SEQ ID No	Start	End	Size (mer)	Corresponding protein	MHC/HLA binding prediction	
2	2556	2564	9	NSP3	H-2-Db	pNSA
4	2586	2593	8	NSP3	HLA-B*35:01	
6	2594	2604	11	NSP3	HLA-C*02:02	
8	2782	2796	15	NSP4	HLA-A*68:02 + HLA-C*02:02	
10	2889	2897	9	NSP4	HLA-A*01:01	pNSB
12	2930	2938	9	NSP4	HLA-B*51:01	
14	3064	3072	9	NSP4	HLA-B*08:01	
16	3127	3144	18	NSP4	H-2-Db + HLA-A*32:01 + 01	
					HLA-B*35:01	
18	3164	3172	9	NSP4	HLA-A*32:01	
20	3179	3187	9	NSP4	HLA-A*65:02	
22	3190	3198	9	NSP4	HLA-B*08:01	
24	3202	3210	9	NSP4	H-2-Db	
26	3361	3369	9	NSP5/3C-like proteinase	HLA-B*08:01	
28	3420	3428	9	NSP5/3C-like proteinase	H-2-Db	pNSB
30	3437	3445	9	NSP5/3C-like proteinase	HLA-A*01:01	
32	3468	3476	9	NSP5/3C-like proteinase	HLA-B*51:01	
34	3612	3634	23	NSP6	HLA-B*35:01 + HLA-A*68:02 + HLA-B*08:01	
36	3641	3663	23	NSP6	01 HLA-B*35:01 + H-2-Db + HLA-B*51:01	
38	4625	4633	9	RpRd/NSP12	H-2-Db	
40	4724	4734	11	RpRd/NSP12	HLA-A*32:01 + HLA-C*02:02	
42	4764	4772	9	RpRd/NSP12	HLA-B*35:01	

TABLE 1-continued

Peptides selected in PR1ab of SARS-CoV-2 (GeneBank: MT126808.1)					
SEQ ID No	Start	End	Size (mer)	Corresponding protein	MHC/HLA binding prediction
44	4789	4799	11	RpRd/NSP12	HLA-A*68:02 + H-2-Db
46	4842	4850	9	RpRd/NSP12	HLA-A*01:01
48	4863	4871	9	RpRd/NSP12	HLA-C*02:02
50	4905	4913	9	RpRd/NSP12	HLA-A*65:02
52	4920	4934	15	RpRd/NSP12	HLA-A*65:02 + HLA-C*02:02
54	4961	4969	9	RpRd/NSP12	HLA-A*65:02
56	5016	5024	9	RpRd/NSP12	H-2-Db
58	5039	5047	9	RpRd/NSP12	HLA-B*08:01
60	5130	5148	19	RpRd/NSP12	HLA-A*01:01 + H-2-Db + HLA-C*02:02 + HLA-B*08:01
62	5221	5229	9	RpRd/NSP12	HLA-B*51:01
64	5246	5254	9	RpRd/NSP12	HLA-B*08:01 + H-2-Db
66	5274	5282	9	RpRd/NSP12	HLA-A*65:02
68	5299	5307	9	RpRd/NSP12	HLA-A*01:01
70	5381	5389	9	Helicase/NSP13	HLA-A*01:01
72	5392	5400	9	Helicase/NSP13	HLA-A*65:02
74	5405	5413	9	Helicase/NSP13	HLA-B*51:01
76	5455	5463	9	Helicase/NSP13	HLA-A*65:02
78	5467	5475	9	Helicase/NSP13	HLA-A*68:02
80	5516	5524	9	Helicase/NSP13	HLA-A*32:01
82	5533	5541	9	Helicase/NSP13	HLA-C*02:02
84	5557	5565	9	Helicase/NSP13	HLA-B*51:01
86	5596	5604	9	Helicase/NSP13	H-2-Db
88	5615	5623	9	Helicase/NSP13	HLA-B*35:01
90	5630	5638	9	Helicase/NSP13	HLA-A*68:02
92	5649	5657	9	Helicase/NSP13	HLA-B*51:01

Example 2—Evaluation of gDPOLYEP Expression In Vitro

[0137] Vaccine plasmids were validated by transfection in eukaryotic cells followed by immunodetection of target proteins. For this, HEK293-T cells were cultured (DMEM+10% SFB) in 24-well plates (10x5 cells/well) for 24 hours (37° C., 5% CO₂). After reaching confluence of 70-80%, the cells were transfected with the recombinant plasmids (pgDNA, pgDNSB, and pgDPOLYEP) or controls (pCDNA3.1) using the Lipofectamine 2000 Kit (Thermo Fisher), according to the manufacturer's instructions. After 24-48 hours of transfection, the cells were trypsinized, washed with PBS solution (twice) and fixed/permeated with Cytofix buffer (BD Biosciences) according to the manufacturer's instructions. After a new washing cycle, the cells were incubated (30 min. on ice) with previously diluted monoclonal anti-gD antibody. After the washing cycle, the cells were incubated with anti-IgG mouse/human antibody conjugated to fluorochrome AlexaFluor 488 (Invitrogen) for 30 min. on ice. After a new washing cycle, the cells were resuspended in PBS+2% SFB and evaluated by flow cytometry in a BD LSRFortessa™ (BD Biosciences) equipment. Data obtained was analyzed using FlowJo v.10 software (TreeStar, OR, USA). As shown in FIG. 3a, from the immunodetection of gD it was possible to determine the in vitro expression of the vaccine antigen gDPOLYEP, as a result of the increase in the frequency of cells marked as positive for gD expression.

[0138] The detection of target protein expression in transfected cells was also evaluated by the immunofluorescence

technique. For this purpose, transfected cells as described above were directly fixed with 4% paraformaldehyde (PFA) solution in PBS (300 µl/well) for 15 minutes at room temperature (RT). After fixing, the cells were permeated with 0.1% Triton X-100 solution in PBS (300 µl/well) for 10 minutes at RT. Then, the cells were washed twice with PBS and blocked with a 2% BSA solution in PBS (30 min., RT). After the blocking step, anti-gD mAb or hyperimmune serum previously diluted in a blocking solution (200 µl/well) were added to the wells. After 1 hour, the wells were washed (3 times) with PBS, the cells were incubated (45 min., RT) with mouse/human anti-IgG antibody conjugated with AlexaFluor 488 (Invitrogen), under agitation. After a new washing cycle, the cells were incubated (20 min., RT) with Hoechst 33342 nucleus stain (Life Technologies) diluted (1/500) in PBS (200 µL/well). After a new washing cycle, the cells were visualized using an Evos FL (Thermo Fisher Scientific) immunofluorescence microscope and the images were captured with magnifications of 100x and 200x. As shown in FIG. 3b, from the immunodetection of gD followed by secondary marking with FITC, it was possible to determine the in vitro expression of the vaccine antigen gDPOLYEP, seen by marking the cells in green in the viewed fields.

Example 3—In Vivo Evaluation of the Vaccines' Immunogenicity

[0139] Animal experiments were conducted according to the Ethical Principles of Animal Experimentation established by the Ethics Committee in Animal Experimentation

[Comitê de Ética em Experimentação Animal (CEUA)]. Mice of the C57BL/6 strain were inoculated intramuscularly according to the following immunization groups: 1) control plasmid (pCDNA3.1); 2) combined pgDNSA and pgDNSB plasmids; 3) polyepitope plasmid (pgDPOLYEP). Each group of animals (n=5-10) received two doses (50 ug/animal) of the vaccine formulations, with a 2-week interval between the doses being adopted. Immediately after the vaccines administration, the animals were electroporated at the application site, with 2 electric pulses of 45 V each being applied, with a duration interval of 450 ms (pulses that form pores in the cell membrane), and 4 pulses of 20 V each, with a duration of 450 ms (transfer pulses), using the NEPA21 SuperElectroporator equipment (NepaGeneCo., Ltd.; Chiba, Japan).

[0140] Intracellular cytokines (ICS) were detected in splenocytes of immunized mice 14 days after the administration of the last vaccine dose. To obtain splenocytes, the spleen of immunized mice was collected after euthanasia and macerated to obtain a cell suspension. The cells obtained were treated for 5 minutes on ice with Ack Lysing Buffer (BioSource International) until red blood cells rupture and then centrifuged at 1500 rpm for 5 minutes. After 2 washing cycles with RPMI medium, the splenocytes obtained were cultured at a ratio of 106 cells/well for 2 hours (37° C., 5% CO₂) in the presence of antigen-specific stimulation (peptides: pep4-VSFCYMHM (SEQ ID NO: 28); pep5-VAY-FNMVYM (SEQ ID NO: 160)) followed by the addition of Brefeldin A (GolgiPlug; BD Biosciences) and stimulation for additional 4 hours. The cells were incubated with RPMI medium (negative control) or in combination with PMA/ionomycin (positive control). After the stimulation period, the cells were washed with PBS (2×), followed by labeling with fluorophore conjugated with antibodies against CD3 and murine (clone 145-2C11, Tonbo), CD8a (clone 53-6.7, BioLegend), CD11a (clone M17/4, eBioscience), CD49d (clone R1-2, eBioscience), all used in dilution of 1:200. The cells were then fixed and permeated with Cytotfix/Cytoperm (BD Bioscience) and stained with monoclonals conjugated with fluorophore against mouse IFN γ (clone XMGI.2, Tonbo) and TNF (clone MP6-XT22, eBioscience). Data was collected in a BD LSRFortessa™ equipment and analyzed using FlowJo software. The cytokine IFN γ secreted by splenocytes of immunized animals was detected using the ELISPOT technique. Briefly, the splenocytes of the immunized animals were stimulated in vitro analogously to that previously described, but without the addition of Brefeldin A.

[0141] In order to characterize the response of T cells induced after vaccination, spleen cells were collected and obtained using the techniques described above. As shown in FIG. 4, after two intramuscular doses of the pgDPOLYEP DNA vaccine (50 μ g), the CD8⁺ T cells vaccine-specific response was evaluated by ICS assay using 2 peptides restricted to H2-Db, contained in the DNA vaccine. The number of spleen cells able to produce IFN γ was deter-

mined, as well as the IFN γ and TNF production frequencies by CD8⁺ T cells activated and experimented with antigen (CD49d⁺CD11a^{hi}CD8 α ^{lo}). Notably, the CD8⁺ T cells of mice that received the pgDPOLYEP DNA vaccine exhibited CD8⁺ T cells with a dominant Th1 phenotype once stimulated with pep, with significantly higher numbers of IFN γ -, TNF- α and IL-2-producing cells, once compared to the other DNA vaccines containing the NSA and NSB portion or the pcDNA3.1 vector (FIG. 4a-f).

[0142] In order to detect IFN γ -producing cells, the spleen cells of the animals after two intramuscular doses of DNA vaccines were subjected to the ELISPOT assay (FIG. 5). As shown, the formulation composed of pgDPOLYEP induced the highest production of IFN γ when compared to the formulations where pgDNSA+pgDNSB or the empty vaccine vector pDNA3.1 was administered. Altogether, the results show that pgDPOLYEP is immunogenic capable of inducing cell-type immune response by inducing IFN γ - and TNF-producing CD8⁺ T lymphocytes.

[0143] Those skilled in the art will immediately understand that the invention is similarly applicable to other vaccine vehicles, that is, the use of the invention is not limited to the incorporation of the sequences encoding the fusion protein by a DNA-type vector. Examples include: the incorporation of the chimeric nucleotide sequence of the invention to vaccine viral vectors, such as those of adenovirus currently used (Janssen, Oxford) for SARS-COV-2 vaccines; the incorporation of the chimeric nucleotide sequence of the invention to microorganisms that express the fusion protein of the invention, for the subsequent inoculation of said preferably purified protein; the use of the chimeric RNA sequence of the invention with RNA vaccine vectors, such as those currently used (Moderna, Pfizer) for SARS-COV-2 vaccines.

[0144] The inventive concept now disclosed and exemplified in one or more ways was treated as an industrial secret and has not been previously disclosed until this patent application filing date or its priority. This industrial secret is immaterial assets of the applicant. The possible future publication of the patent application does not constitute, in itself, an authorization for use by third parties, being only useful as: (i) third-party acknowledgment of the existence of said industrial secret on the filing date; (ii) unequivocal indication of its holder; and (iii) stimulation to the development of new improvements from the concept now disclosed in order to avoid reinvestment in the development of the same asset already held by the applicant.

[0145] It is immediately warned that any commercial use requires the holder's authorization and that the unauthorized use entails sanctions provided for in the Law. In this context, it is clarified that from the disclosure of the present inventive concept, those skilled in the art may consider other ways of realizing the invention which are not identical to those merely exemplified above, but that in the event of commercial use intention, such forms may be considered as being within the scope of the attached claims.

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<400> SEQUENCE: 59

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<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 60

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Ser Met Met

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<400> SEQUENCE: 61

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<400> SEQUENCE: 62

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atggtgatgt tcaccccctt ggtgcccttc tggatcacca tcgcctacgg cggcggcagc	360
aggggtggtg tcaacggcgt gagcttcggc ggcggcagcg ccctgtgcac ctctctgctg	420
aacaagggcg gcggcagcta cctgaagctg aggagcgcgc tgctggggcg cggcagcacc	480
cagtacaaca ggtacctggc cctggggcgc ggcagcaccc ccaagtacaa gtctgtgagg	540
atcggcggcg gcagcgtgag cttctgtctac atgcaccaca tgggcggcgcg cagcggcacc	600
gacctggagg gcaacttcta cggcggcggc agcctggcct ggctgtacgc cgccgtgatc	660
ggcgggcgga gcctgccctt cgccatgggc atcatcgcca tgagcgccct cgccatgatg	720
ttcgtgaagc acaagcacgc cggcggcggc agcctgccc gcctggccac cgtggcctac	780
ttcaacatgg tgtacatgcc cgccagctgg gtgatgagga tcggcgggcg cagcgtggtg	840
gacagctact acagcctgct gggcggcggc agcaagatct tcgtggacgg cgtgcccttc	900
gtggtgggcg gcggcagcct ggtgtacgcc gccgaccccg ccatggggcg cggcagcagc	960
gtggccgccc tgaccaacaa cgtggccttc ggcggcgcca gcatcagcga ctacgactac	1020
tacaggtagc gcggcggcag cttcgtggtg gaggtggtgg acaagtacgg cggcggcagc	1080
aggctgtact acgacagcat gagctacggc ggcggcagct tcgcctacac caagaggaac	1140
gtgateccca ccatcaccca gatgggcggc ggcagcaggc agttccacca gaagctgctg	1200
aaggcgggcg gcagcaggcg catgcccac atgctgagga tcggcgggcg cagcagcctg	1260
agccacaggt tctacaggct gggcggcggc agcgacacgg acttcgtgaa cgagttctac	1320
gcctacctga ggaagcactt cagcatgatg ggcggcgcca gcctgcccta ccccacccc	1380
agcaggatcg gcggcggcag cctgatgatc gagaggttcg tgagcctggg cggcggcagc	1440
cacctgtacc tgcagtacat caggaaggcg ggcggcagcc tgaccaacga caacaccagc	1500
aggtagggcg gcggcagcgt gaccgacgtg acccagctgt acctggggcg cggcagcatg	1560

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agctactact gcaagagcca caagggcggc ggcagcttcc cctgtgctgc caacggccag 1620
gtgggcggcg gcagcaagct gttcgccgcc gagaccctga agggcggcgg cagcgagacc 1680
ttcaagctga gctacggcat cggcggcggc agcaaggtgc agatcggcga gtacaccttc 1740
ggcggcggca gcgtggtgta caggggcacc accacctacg gcggcggcag catgccctg 1800
agcgcctccc ccttggtggg cggcggcagc gtgggcacgc agaagtacag caccctgggc 1860
ggcggcagct tcgccatcgg cctggccctg tactacggcg gcggcagcta caccgcctgc 1920
agccacgccg ccgtggggcg cggcagcctg cccatcgaca agtgcagcag gatc 1974

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<210> SEQ ID NO 96
<211> LENGTH: 658
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Peptide sequence encoding the polyepitope

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<400> SEQUENCE: 96

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Ser Ala Ser Val Tyr Tyr Ser Gln Leu Gly Gly Gly Ser Val Ala Val
1           5           10          15

Lys Met Phe Asp Ala Gly Gly Gly Ser Tyr Val Asn Thr Phe Ser Ser
20          25          30

Thr Phe Asn Val Gly Gly Gly Ser Phe Val Ala Ala Ile Phe Tyr Leu
35          40          45

Ile Thr Pro Val His Val Met Gly Gly Gly Ser Phe Ser Ala Val Gly
50          55          60

Asn Ile Cys Tyr Gly Gly Gly Ser Val Pro Tyr Cys Tyr Asp Thr Asn
65          70          75          80

Val Gly Gly Gly Ser Phe Met Arg Phe Arg Arg Ala Phe Gly Gly Gly
85          90          95

Gly Ser Gln Trp Met Val Met Phe Thr Pro Leu Val Pro Phe Trp Ile
100         105         110

Thr Ile Ala Tyr Gly Gly Gly Ser Arg Val Val Phe Asn Gly Val Ser
115         120         125

Phe Gly Gly Gly Ser Ala Leu Cys Thr Phe Leu Leu Asn Lys Gly Gly
130         135         140

Gly Ser Tyr Leu Lys Leu Arg Ser Asp Val Leu Gly Gly Gly Ser Thr
145         150         155         160

Gln Tyr Asn Arg Tyr Leu Ala Leu Gly Gly Gly Ser Thr Pro Lys Tyr
165         170         175

Lys Phe Val Arg Ile Gly Gly Gly Ser Val Ser Phe Cys Tyr Met His
180         185         190

His Met Gly Gly Gly Ser Gly Thr Asp Leu Glu Gly Asn Phe Tyr Gly
195         200         205

Gly Gly Ser Leu Ala Trp Leu Tyr Ala Ala Val Ile Gly Gly Gly Ser
210         215         220

Leu Pro Phe Ala Met Gly Ile Ile Ala Met Ser Ala Phe Ala Met Met
225         230         235         240

Phe Val Lys His Lys His Ala Gly Gly Gly Ser Leu Pro Ser Leu Ala
245         250         255

Thr Val Ala Tyr Phe Asn Met Val Tyr Met Pro Ala Ser Trp Val Met
260         265         270

Arg Ile Gly Gly Gly Ser Val Val Asp Ser Tyr Tyr Ser Leu Leu Gly

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275					280					285					
Gly	Gly	Ser	Lys	Ile	Phe	Val	Asp	Gly	Val	Pro	Phe	Val	Val	Gly	Gly
290						295					300				
Gly	Ser	Leu	Val	Tyr	Ala	Ala	Asp	Pro	Ala	Met	Gly	Gly	Gly	Ser	Ser
305					310					315				320	
Val	Ala	Ala	Leu	Thr	Asn	Asn	Val	Ala	Phe	Gly	Gly	Gly	Ser	Ile	Ser
				325					330					335	
Asp	Tyr	Asp	Tyr	Tyr	Arg	Tyr	Gly	Gly	Gly	Ser	Phe	Val	Val	Glu	Val
			340					345					350		
Val	Asp	Lys	Tyr	Gly	Gly	Gly	Ser	Arg	Leu	Tyr	Tyr	Asp	Ser	Met	Ser
		355					360					365			
Tyr	Gly	Gly	Gly	Ser	Phe	Ala	Tyr	Thr	Lys	Arg	Asn	Val	Ile	Pro	Thr
370						375					380				
Ile	Thr	Gln	Met	Gly	Gly	Gly	Ser	Arg	Gln	Phe	His	Gln	Lys	Leu	Leu
385					390					395				400	
Lys	Gly	Gly	Gly	Ser	Arg	Ala	Met	Pro	Asn	Met	Leu	Arg	Ile	Gly	Gly
				405					410					415	
Gly	Ser	Ser	Leu	Ser	His	Arg	Phe	Tyr	Arg	Leu	Gly	Gly	Gly	Ser	Asp
			420					425					430		
Thr	Asp	Phe	Val	Asn	Glu	Phe	Tyr	Ala	Tyr	Leu	Arg	Lys	His	Phe	Ser
		435					440					445			
Met	Met	Gly	Gly	Gly	Ser	Leu	Pro	Tyr	Pro	Asp	Pro	Ser	Arg	Ile	Gly
450						455					460				
Gly	Gly	Ser	Leu	Met	Ile	Glu	Arg	Phe	Val	Ser	Leu	Gly	Gly	Gly	Ser
465					470					475				480	
His	Leu	Tyr	Leu	Gln	Tyr	Ile	Arg	Lys	Gly	Gly	Gly	Ser	Leu	Thr	Asn
			485					490						495	
Asp	Asn	Thr	Ser	Arg	Tyr	Gly	Gly	Gly	Ser	Val	Thr	Asp	Val	Thr	Gln
			500					505					510		
Leu	Tyr	Leu	Gly	Gly	Gly	Ser	Met	Ser	Tyr	Tyr	Cys	Lys	Ser	His	Lys
		515					520					525			
Gly	Gly	Gly	Ser	Phe	Pro	Leu	Cys	Ala	Asn	Gly	Gln	Val	Gly	Gly	Gly
		530				535					540				
Ser	Lys	Leu	Phe	Ala	Ala	Glu	Thr	Leu	Lys	Gly	Gly	Gly	Ser	Glu	Thr
545				550						555				560	
Phe	Lys	Leu	Ser	Tyr	Gly	Ile	Gly	Gly	Gly	Ser	Lys	Val	Gln	Ile	Gly
			565					570						575	
Glu	Tyr	Thr	Phe	Gly	Gly	Gly	Ser	Val	Val	Tyr	Arg	Gly	Thr	Thr	Thr
			580					585					590		
Tyr	Gly	Gly	Gly	Ser	Met	Pro	Leu	Ser	Ala	Pro	Thr	Leu	Val	Gly	Gly
		595				600						605			
Gly	Ser	Val	Gly	Met	Gln	Lys	Tyr	Ser	Thr	Leu	Gly	Gly	Gly	Ser	Phe
		610				615					620				
Ala	Ile	Gly	Leu	Ala	Leu	Tyr	Tyr	Gly	Gly	Gly	Ser	Tyr	Thr	Ala	Cys
625					630					635				640	
Ser	His	Ala	Ala	Val	Gly	Gly	Gly	Ser	Leu	Pro	Ile	Asp	Lys	Cys	Ser
			645					650					655		
Arg Ile															

<210> SEQ ID NO 97

<211> LENGTH: 819

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Nucleotide sequence coding for the modified
    form of herpes simplex type-1 (HSV-1) glycoprotein D (gD) - gD
    fragment in the N-terminal portion

<400> SEQUENCE: 97

aagcttacca tgatgggagg ggctgctgcc agactgggag ctgtgattct gttcgtggtc      60
attgtggggc tgcattggagt gcgggggaaa tacgctctgg ccgatgcaag cctgaaaatg      120
gcagacccta acagattcag agggaaagac ctgccctgtc tggatcagct gactgatccc      180
ccaggcgtga gacgggtgta tcatatccaa gctggcctgc cagatccatt ccagccccc      240
tccctgccta tcaccgtcta ctacgcagtg ctggagagag cctgtaggtc cgtgctgctg      300
aatgctccta gcgaagcccc tcaaattgtg cggggcgccct ccgaagacgt gagaaagcag      360
ccatataatc tgaccatcgc ttggtttagg atggggcgca actgcgccat tcctatcacc      420
gtgatggagt atactgaatg cagctataac aagtcctcgg gcgcctgccc aattagaacc      480
caacctaggt ggaactatta tgacagcttt tccgcagtcg gcgaagacaa cctgggggtc      540
ctgatgcacg ccccgctttt tgaaactgcc gggacatatc tgcggctggt caaatcaac      600
gattggacag aaatcaccca gtccatcctg gaacataggg ccaaaggag ctgcaaatat      660
gcccctgccac tgaggatccc ccttccgct tgtctgtccc ctcaagccta ccagcaagga      720
gtgacagtgg acagcatcgg catgctgccc agattcattc ccgagaacca gagaactgtg      780
gctgtgtact ccctgaagat tgccggatgg cacggatcc      819

<210> SEQ ID NO 98
<211> LENGTH: 270
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Amino acid sequence of the modified form of
    herpes simplex type-1 (HSV-1) gD - gD fragment in the N-terminal
    portion

<400> SEQUENCE: 98

Met Met Gly Gly Ala Ala Ala Arg Leu Gly Ala Val Ile Leu Phe Val
1          5          10         15
Val Ile Val Gly Leu His Gly Val Arg Gly Lys Tyr Ala Leu Ala Asp
20         25         30
Ala Ser Leu Lys Met Ala Asp Pro Asn Arg Phe Arg Gly Lys Asp Leu
35         40         45
Pro Val Leu Asp Gln Leu Thr Asp Pro Pro Gly Val Arg Arg Val Tyr
50         55         60
His Ile Gln Ala Gly Leu Pro Asp Pro Phe Gln Pro Pro Ser Leu Pro
65         70         75         80
Ile Thr Val Tyr Tyr Ala Val Leu Glu Arg Ala Cys Arg Ser Val Leu
85         90         95
Leu Asn Ala Pro Ser Glu Ala Pro Gln Ile Val Arg Gly Ala Ser Glu
100        105        110
Asp Val Arg Lys Gln Pro Tyr Asn Leu Thr Ile Ala Trp Phe Arg Met
115        120        125
Gly Gly Asn Cys Ala Ile Pro Ile Thr Val Met Glu Tyr Thr Glu Cys
130        135        140

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Ser	Tyr	Asn	Lys	Ser	Leu	Gly	Ala	Cys	Pro	Ile	Arg	Thr	Gln	Pro	Arg
145					150					155					160
Trp	Asn	Tyr	Tyr	Asp	Ser	Phe	Ser	Ala	Val	Ser	Glu	Asp	Asn	Leu	Gly
			165						170					175	
Phe	Leu	Met	His	Ala	Pro	Ala	Phe	Glu	Thr	Ala	Gly	Thr	Tyr	Leu	Arg
			180					185					190		
Leu	Val	Lys	Ile	Asn	Asp	Trp	Thr	Glu	Ile	Thr	Gln	Phe	Ile	Leu	Glu
		195					200					205			
His	Arg	Ala	Lys	Gly	Ser	Cys	Lys	Tyr	Ala	Leu	Pro	Leu	Arg	Ile	Pro
	210					215					220				
Pro	Ser	Ala	Cys	Leu	Ser	Pro	Gln	Ala	Tyr	Gln	Gln	Gly	Val	Thr	Val
225					230					235					240
Asp	Ser	Ile	Gly	Met	Leu	Pro	Arg	Phe	Ile	Pro	Glu	Asn	Gln	Arg	Thr
			245					250						255	
Val	Ala	Val	Tyr	Ser	Leu	Lys	Ile	Ala	Gly	Trp	His	Gly	Ser		
			260				265					270			

<210> SEQ ID NO 99
 <211> LENGTH: 396
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Nucleotide sequence coding for the modified form of herpes simplex type-1 (HSV-1) glycoprotein D (gD) - gD fragment in the C-terminal portion

<400> SEQUENCE: 99

ggatcctcta gaaagggccc ttacacttcc acactgctgc cccctgagct gagcgaaaca	60
cccaatgccca ccagagccga gctggctccc gaggaccccg aggacagcgc cctgctggag	120
gaccagtcg gcacagtcgc ccacagatc cctcccaatt ggcacattcc cagcattcag	180
gacgcagcaa cccctacca cccaccagca acccctaata atatgggcct gatcgaggc	240
gcagtcggag gctccctgct ggcagcactg gtcatctgtg ggatcgtcta ctggatgagg	300
cggcggacac agaaggcacc aaagcggatt cggctgccac acattcggga agacgatcag	360
ccttcctccc accagccact gttttactaa ctcgag	396

<210> SEQ ID NO 100
 <211> LENGTH: 129
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Amino acid sequence encoding the modified form of herpes simplex type-1 (HSV-1) glycoprotein D (gD) - gD fragment in the C-terminal portion

<400> SEQUENCE: 100

Gly	Ser	Ser	Arg	Lys	Ala	Pro	Tyr	Thr	Ser	Thr	Leu	Leu	Pro	Pro	Glu
1				5					10					15	
Leu	Ser	Glu	Thr	Pro	Asn	Ala	Thr	Gln	Pro	Glu	Leu	Ala	Pro	Glu	Asp
			20					25					30		
Pro	Glu	Asp	Ser	Ala	Leu	Leu	Glu	Asp	Pro	Val	Gly	Thr	Val	Ala	Pro
		35					40					45			
Gln	Ile	Pro	Pro	Asn	Trp	His	Ile	Pro	Ser	Ile	Gln	Asp	Ala	Ala	Thr
	50					55					60				
Pro	Tyr	His	Pro	Pro	Ala	Thr	Pro	Asn	Asn	Met	Gly	Leu	Ile	Ala	Gly
65					70					75					80

Tyr

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<210> SEQ ID NO 101
<211> LENGTH: 3189
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Covid19_gD_POLIEP (DNA) (5'-->3')
```

<400> SEQUENCE: 101

aagcttacc	tgatgggagg	ggctgctgcc	agactgggag	ctgtgattct	gttcgtggtc	60
attgtggggc	tgcattggagt	gcgggggaaa	tacgctctgg	ccgatgcaag	cctgaaaatg	120
gcagacccta	acagattcag	agggaaagac	ctgccctgtc	tggatcagct	gactgatccc	180
ccaggcggtg	gacgggtgta	tcatatccaa	gtgggectgc	cagatccatt	ccagccccc	240
tccttgcccta	tcaccgtcta	ctacgcagtg	ctggagagag	cctgtaggtc	cgtgctgctg	300
aatgctccta	gcgaagcccc	tcaaattgtg	cgggggcgct	ccgaagacgt	gagaaagcag	360
ccatataatc	tgaccatcgc	ttggtttagg	atggggcgga	actgcgccat	tcctatcacc	420
gtgatggagt	atactgaatg	cagctataac	aagtccctgg	gcgcctgcc	aattagaacc	480
caacctaggt	ggaactatta	tgacagcttt	tccgcagtc	gcgaagacaa	cctgggggtc	540
ctgatgcacg	ccccgccttt	tgaaactgcc	gggacatata	tgcggtcgtg	caaaatcaac	600
gattggacag	aaatcaccca	gttcatcctg	gaacataggg	ccaaagggag	ctgcaaatat	660
gccttgccc	tgaggatccc	cccttcctgt	tgtctgtccc	ctcaagccta	ccagcaagga	720
gtgacagtgg	acagcatcgg	catgctgccc	agattcattc	ccgagaacca	gagaactgtg	780
gctgtgtact	ccctgaagat	tgccggatgg	cacggatcca	gcgccagcgt	gtactacagc	840
cagctggggc	gcggcagcgt	ggccgtgaag	atgttcgacg	ccggcggcgg	cagctacgtg	900
aacaccttca	gcagcacctt	caacgtgggc	ggcggcagct	tcgtggccgc	catcttctac	960
ctgatcacc	ccgtgcacgt	gatggggcgc	ggcagcttca	gcgccgtggg	caacatctgc	1020
tacggcggcg	gcagcgtgcc	ctactgtctc	gacaccaacg	tgggcggcgg	cagcttcatg	1080
aggttcagga	gggccttcgg	cggcggcggc	agccagtgga	tggtgatgtt	cacccccctg	1140
gtgcccttct	ggatcaccat	cgctacggc	ggcggcagca	gggtggtgtt	caacggcgtg	1200
agcttcggcg	gcggcagcgc	cctgtgcacc	ttcctgctga	acaaggcgcg	cggcagctac	1260
ctgaagctga	ggagcgacgt	gctggggcgc	ggcagcacc	agtacaacag	gtacctggcc	1320
ctggggcggc	gcagcacccc	caagtacaag	ttcgtgagga	tcggcggcgg	cagcgtgagc	1380
ttctgtctaca	tgcaccacat	gggcggcggc	agcggcaccc	acctggaggg	caacttctac	1440
ggcggcggca	gcctggcctg	gctgtacgcc	gccgtgatcg	gcggcggcag	cctgcccttc	1500
gccatgggca	tcatgcccat	gagcgccttc	gccatgatgt	tcgtgaagca	caagcacgcc	1560
ggcggcggca	gcctgcccag	cctggccacc	gtggcctact	tcaacatggt	gtacatgccc	1620

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gccagctggg tgatgaggat cggcggcggc agcgtggtgg acagctacta cagcctgctg 1680
ggcggcggca gcaagatctt cgtggacggc gtgcccttcg tggtagggcg cggcagcctg 1740
gtgtacgccg ccgaccccg catggcgggc ggcagcagcg tggccgcctt gaccaacaac 1800
gtggccttcg gcgcgggcag catcagcgac tacgactact acaggtacgg cggcggcagc 1860
ttcgtggtgg aggtggtgga caagtacggc ggcggcagca ggctgtacta cgacagcatg 1920
agctacggcg cggcagcgtt cgcctacacc aagaggaacg tgatccccac catcaccag 1980
atggcgggcg gcagcaggca gtccaccag aagctgctga agggcggcgg cagcagggcc 2040
atgcccaaca tgctgaggat cggcggcggc agcagcctga gccacaggtt ctacaggctg 2100
ggcggcggca cgcacaccga cttcgtgaac gagttctacg cctacctgag gaagcacttc 2160
agcatgatgg cggcgggcag cctgccctac cccgacccca gcaggatcgg cggcggcagc 2220
ctgatgatcg agaggttcgt gagcctgggc ggcggcagcc acctgtacct gcagtacatc 2280
aggaaggggc cggcagcctt gaccaacgac aacaccagca ggtacggcgg cggcagcgtg 2340
accgacgtga cccagctgta cctggcgggc ggcagcatga gctactactg caagagccac 2400
aaggcgggcg gcagcttccc cctgtgcgcc aacggcccagg tggcgggcgg cagcaagctg 2460
ttcgccgccg agacctgaa gggcgggcggc agcgagacct tcaagctgag ctacggcatc 2520
ggcggcggca gcaaggtgca gatcggcgag tacaccttcg cgcggcgagc cgtggtgtac 2580
aggggcacca ccacctacgg cggcggcagc atgcccctga gcgccccac cctggtgggc 2640
ggcggcagcg tgggcatgca gaagtacagc accctggcg cggcagcgtt cgccatcggc 2700
ctggccctgt actacggcgg cggcagctac accgcctgca gccacgcgcg cgtggcgggc 2760
ggcagcctgc ccatcgacaa gtgcagcagg atcggtacct ctagaaaggc cccttacct 2820
tccacactgc tgccccctga gctgagcgaa acacccaatg ccaccagcc cgagctggct 2880
cccgaggacc ccgaggacag cgccctgctg gaggaccagc tcggcacagt cgccccacag 2940
atccctccca attggcact tcccagcatt caggacgcag caacccccta ccaccacca 3000
gcaaccccta ataatatggg cctgatcgca ggcgcagtcg gaggtccct gctggcagca 3060
ctggtcatct gtgggatcgt ctactggatg aggcggcgga cacagaaggc accaaagcgg 3120
attcggtgct cacacattcg ggaagacgat cagccttcct cccaccagcc actgttttac 3180
taactcgag 3189

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<210> SEQ ID NO 102

<211> LENGTH: 1057

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Covid19_gD_POLIEP (protein)

<400> SEQUENCE: 102

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Met Met Gly Gly Ala Ala Ala Arg Leu Gly Ala Val Ile Leu Phe Val
1           5           10          15

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Val Ile Val Gly Leu His Gly Val Arg Gly Lys Tyr Ala Leu Ala Asp
20          25          30

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Ala Ser Leu Lys Met Ala Asp Pro Asn Arg Phe Arg Gly Lys Asp Leu
35          40          45

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Pro Val Leu Asp Gln Leu Thr Asp Pro Pro Gly Val Arg Arg Val Tyr
50          55          60

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His 65	Ile	Gln	Ala	Gly	Leu 70	Pro	Asp	Pro	Phe	Gln 75	Pro	Pro	Ser	Leu 80	Pro
Ile	Thr	Val	Tyr	Tyr	Ala	Val	Leu	Glu	Arg	Ala	Cys	Arg	Ser	Val	Leu
				85					90					95	
Leu	Asn	Ala	Pro	Ser	Glu	Ala	Pro	Gln	Ile	Val	Arg	Gly	Ala	Ser	Glu
			100					105					110		
Asp	Val	Arg	Lys	Gln	Pro	Tyr	Asn	Leu	Thr	Ile	Ala	Trp	Phe	Arg	Met
		115					120					125			
Gly	Gly	Asn	Cys	Ala	Ile	Pro	Ile	Thr	Val	Met	Glu	Tyr	Thr	Glu	Cys
	130					135					140				
Ser	Tyr	Asn	Lys	Ser	Leu	Gly	Ala	Cys	Pro	Ile	Arg	Thr	Gln	Pro	Arg
145					150					155					160
Trp	Asn	Tyr	Tyr	Asp	Ser	Phe	Ser	Ala	Val	Ser	Glu	Asp	Asn	Leu	Gly
				165					170					175	
Phe	Leu	Met	His	Ala	Pro	Ala	Phe	Glu	Thr	Ala	Gly	Thr	Tyr	Leu	Arg
			180					185					190		
Leu	Val	Lys	Ile	Asn	Asp	Trp	Thr	Glu	Ile	Thr	Gln	Phe	Ile	Leu	Glu
		195					200					205			
His	Arg	Ala	Lys	Gly	Ser	Cys	Lys	Tyr	Ala	Leu	Pro	Leu	Arg	Ile	Pro
	210					215					220				
Pro	Ser	Ala	Cys	Leu	Ser	Pro	Gln	Ala	Tyr	Gln	Gln	Gly	Val	Thr	Val
225					230					235					240
Asp	Ser	Ile	Gly	Met	Leu	Pro	Arg	Phe	Ile	Pro	Glu	Asn	Gln	Arg	Thr
				245					250					255	
Val	Ala	Val	Tyr	Ser	Leu	Lys	Ile	Ala	Gly	Trp	His	Gly	Ser	Ser	Ala
				260				265					270		
Ser	Val	Tyr	Tyr	Ser	Gln	Leu	Gly	Gly	Gly	Ser	Val	Ala	Val	Lys	Met
		275					280					285			
Phe	Asp	Ala	Gly	Gly	Gly	Ser	Tyr	Val	Asn	Thr	Phe	Ser	Ser	Thr	Phe
	290					295					300				
Asn	Val	Gly	Gly	Gly	Ser	Phe	Val	Ala	Ala	Ile	Phe	Tyr	Leu	Ile	Thr
305					310					315					320
Pro	Val	His	Val	Met	Gly	Gly	Gly	Ser	Phe	Ser	Ala	Val	Gly	Asn	Ile
				325					330					335	
Cys	Tyr	Gly	Gly	Gly	Ser	Val	Pro	Tyr	Cys	Tyr	Asp	Thr	Asn	Val	Gly
			340					345					350		
Gly	Gly	Ser	Phe	Met	Arg	Phe	Arg	Arg	Ala	Phe	Gly	Gly	Gly	Gly	Ser
		355					360					365			
Gln	Trp	Met	Val	Met	Phe	Thr	Pro	Leu	Val	Pro	Phe	Trp	Ile	Thr	Ile
	370					375					380				
Ala	Tyr	Gly	Gly	Gly	Ser	Arg	Val	Val	Phe	Asn	Gly	Val	Ser	Phe	Gly
385					390					395					400
Gly	Gly	Ser	Ala	Leu	Cys	Thr	Phe	Leu	Leu	Asn	Lys	Gly	Gly	Gly	Ser
				405				410						415	
Tyr	Leu	Lys	Leu	Arg	Ser	Asp	Val	Leu	Gly	Gly	Gly	Ser	Thr	Gln	Tyr
			420					425					430		
Asn	Arg	Tyr	Leu	Ala	Leu	Gly	Gly	Gly	Ser	Thr	Pro	Lys	Tyr	Lys	Phe
		435				440						445			
Val	Arg	Ile	Gly	Gly	Gly	Ser	Val	Ser	Phe	Cys	Tyr	Met	His	His	Met
	450					455					460				
Gly	Gly	Gly	Ser	Gly	Thr	Asp	Leu	Glu	Gly	Asn	Phe	Tyr	Gly	Gly	Gly

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465		470		475		480
Ser Leu Ala Trp	Leu Tyr Ala Ala Val Ile Gly Gly Gly Ser Leu Pro					
	485			490		495
Phe Ala Met Gly Ile Ile Ala Met Ser Ala Phe Ala Met Met Phe Val						
	500			505		510
Lys His Lys His Ala Gly Gly Gly Ser Leu Pro Ser Leu Ala Thr Val						
	515			520		525
Ala Tyr Phe Asn Met Val Tyr Met Pro Ala Ser Trp Val Met Arg Ile						
	530			535		540
Gly Gly Gly Ser Val Val Asp Ser Tyr Tyr Ser Leu Leu Gly Gly Gly						
545		550		555		560
Ser Lys Ile Phe Val Asp Gly Val Pro Phe Val Val Gly Gly Gly Ser						
	565			570		575
Leu Val Tyr Ala Ala Asp Pro Ala Met Gly Gly Gly Ser Ser Val Ala						
	580			585		590
Ala Leu Thr Asn Asn Val Ala Phe Gly Gly Gly Ser Ile Ser Asp Tyr						
	595			600		605
Asp Tyr Tyr Arg Tyr Gly Gly Gly Ser Phe Val Val Glu Val Val Asp						
	610			615		620
Lys Tyr Gly Gly Gly Ser Arg Leu Tyr Tyr Asp Ser Met Ser Tyr Gly						
625		630		635		640
Gly Gly Ser Phe Ala Tyr Thr Lys Arg Asn Val Ile Pro Thr Ile Thr						
	645			650		655
Gln Met Gly Gly Gly Ser Arg Gln Phe His Gln Lys Leu Leu Lys Gly						
	660			665		670
Gly Gly Ser Arg Ala Met Pro Asn Met Leu Arg Ile Gly Gly Gly Ser						
	675			680		685
Ser Leu Ser His Arg Phe Tyr Arg Leu Gly Gly Gly Ser Asp Thr Asp						
	690			695		700
Phe Val Asn Glu Phe Tyr Ala Tyr Leu Arg Lys His Phe Ser Met Met						
705		710		715		720
Gly Gly Gly Ser Leu Pro Tyr Pro Asp Pro Ser Arg Ile Gly Gly Gly						
	725			730		735
Ser Leu Met Ile Glu Arg Phe Val Ser Leu Gly Gly Gly Ser His Leu						
	740			745		750
Tyr Leu Gln Tyr Ile Arg Lys Gly Gly Gly Ser Leu Thr Asn Asp Asn						
	755			760		765
Thr Ser Arg Tyr Gly Gly Gly Ser Val Thr Asp Val Thr Gln Leu Tyr						
	770			775		780
Leu Gly Gly Gly Ser Met Ser Tyr Tyr Cys Lys Ser His Lys Gly Gly						
785		790		795		800
Gly Ser Phe Pro Leu Cys Ala Asn Gly Gln Val Gly Gly Gly Ser Lys						
	805			810		815
Leu Phe Ala Ala Glu Thr Leu Lys Gly Gly Gly Ser Glu Thr Phe Lys						
	820			825		830
Leu Ser Tyr Gly Ile Gly Gly Gly Ser Lys Val Gln Ile Gly Glu Tyr						
	835			840		845
Thr Phe Gly Gly Gly Ser Val Val Tyr Arg Gly Thr Thr Thr Tyr Gly						
	850			855		860
Gly Gly Ser Met Pro Leu Ser Ala Pro Thr Leu Val Gly Gly Gly Ser						
865		870		875		880

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Val Gly Met Gln Lys Tyr Ser Thr Leu Gly Gly Gly Ser Phe Ala Ile
885 890 895

Gly Leu Ala Leu Tyr Tyr Gly Gly Gly Ser Tyr Thr Ala Cys Ser His
900 905 910

Ala Ala Val Gly Gly Gly Ser Leu Pro Ile Asp Lys Cys Ser Arg Ile
915 920 925

Gly Ser Ser Arg Lys Ala Pro Tyr Thr Ser Thr Leu Leu Pro Pro Glu
930 935 940

Leu Ser Glu Thr Pro Asn Ala Thr Gln Pro Glu Leu Ala Pro Glu Asp
945 950 955 960

Pro Glu Asp Ser Ala Leu Leu Glu Asp Pro Val Gly Thr Val Ala Pro
965 970 975

Gln Ile Pro Pro Asn Trp His Ile Pro Ser Ile Gln Asp Ala Ala Thr
980 985 990

Pro Tyr His Pro Pro Ala Thr Pro Asn Asn Met Gly Leu Ile Ala Gly
995 1000 1005

Ala Val Gly Gly Ser Leu Leu Ala Ala Leu Val Ile Cys Gly Ile
1010 1015 1020

Val Tyr Trp Met Arg Arg Arg Thr Gln Lys Ala Pro Lys Arg Ile
1025 1030 1035

Arg Leu Pro His Ile Arg Glu Asp Asp Gln Pro Ser Ser His Gln
1040 1045 1050

Pro Leu Phe Tyr
1055

<210> SEQ ID NO 103

<211> LENGTH: 4548

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Covid19_gD_NSa (DNA) (5'-->3')

<400> SEQUENCE: 103

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gcagacccta acagattcag agggaaagac ctgcccgtgc tggatcagct gactgatccc	180
ccaggcgtga gacgggtgta tcatatccaa gctggcctgc cagatccatt ccagccccc	240
tcctcgctta tcaccgtcta ctacgcagtg ctggagagag cctgtaggtc cgtgctgctg	300
aatgctccta gcgaagcccc tcaaattgtg cggggcgccct ccgaagacgt gagaaagcag	360
ccatataatc tgaccatgcg ttggttttagg atgggcgcca actgcgccat tcctatcacc	420
gtgatggagt atactgaatg cagctataac aagtccttgg gcgcctgccc aattagaacc	480
caacctaggt ggaactatta tgacagcttt tccgcagtca gcgaagacaa cctgggggttc	540
ctgatgcacg ccccgctttt tgaaaatgcc gggacatatc tgcggctggt caaaatcaac	600
gattggacag aaatcaccca gttcctcctg gaacataggg ccaaaggag ctgcaaatat	660
gccctgccac tgaggatccc ccttccgct tgtctgtccc ctcaagccta ccagcaagga	720
gtgacagtgg acagcatcgg catgctgccc agattcattc ccgagaacca gagaactgtg	780
gctgtgtact ccctgaagat tgccgatgg cacggatcca gcgcgaaaag cgcgagcgtg	840
tattatagcc agctgatgtg ccagccgatt ctgctgctgg atcaggcgct ggtgagcgat	900

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gtggcgata ggcggaagt ggcggtgaaa atgtttgatg cgtatgtgaa caccttttagc	960
agcaccttta acgtgccgat ggaanaactg aaaaccctgg tggcgaccgc ggaagcggaa	1020
ctggcgaaaa acgtgagcct ggataacgtg ctgagcacct ttattagcgc ggcgcgccag	1080
ggcttttggtg atagcgatgt ggaaccacaa gatgtggtgg aatgcctgaa actgagccat	1140
cagagcgata ttgaagtgc cgcgatagc tgcaacaact atatgctgac ctataacaaa	1200
gtgaaaaaca tgaccccgcg cgatctgggc gcgtgcattg attgcagcgc gcgccatatt	1260
aacgcgcagg tggcgaaaag ccataacatt gcgtgtgatt ggaacgtgaa agattttatg	1320
agcctgagcg aacagctgcg caaacagatt cgcagcgcgg cgaaaaaaaa caacctgccg	1380
tttaactga cctgcgcgac caccgcgcag gtggtgaacg tggtgaccac caaattgcg	1440
ctgaaaggcg gcaaaattgt gaacaactgg ctgaaacagc tgattaaagt gacctggtg	1500
tttctgtttg tggcgcgat tttttatctg attaccccg tgcatgtgat gagcaaacat	1560
accgatttta gcagcgaaat tattggctat aaagcgattg atggcgcgct gaccgcgat	1620
attgcgagca cggatactg ctttgcgaac aaacatgcgg attttgatac ctggttttagc	1680
cagcgcggcg gcagctatac caacgataaa gcgtgccgc tgattgcggc ggtgattacc	1740
cgcaagtgg gctttgtggt gccgggacct cgggcacca ttctgcgcac caccaacggc	1800
gattttctgc attttctgcc gcgcgtgttt agcgcggtgg gcaacatttg ctatacccg	1860
agcaactga ttgaatatac cgattttgcg accagcgcgt gcgtgctggc ggcggaatgc	1920
accattttta aagatgcgag cggcaaacgg gtgcccgtatt gctatgatac caacgtgctg	1980
gaaggcagcg tggcgatga aagcctgcgc ccggataccc gctatgtgct gatggatggc	2040
agcattatc agtttccgaa cacctatctg gaaggcagcg tgcgcgtggg gaccacctt	2100
gatagcgaat attgccgcca tggcacctgc gaacgcagcg aagcggcggt gtgcgtgagc	2160
accagcggcc gctgggtgct gaacaacgat tattatcgca gcctgccggg cgtgttttgc	2220
ggcgtggatg cggtgaaact gctgaccaac atgtttaccc cgtgattca gccgattggc	2280
gcgctggata ttagcgcgag cattgtggcg ggcggcattg tggcgattgt ggtgacctgc	2340
ctggcgatatt attttatgcg ctttcgccgc gcgtttggcg aatatagcca tgtggtggcg	2400
tttaaccccc tgctgtttct gatgagcttt accgtgctgt gcctgacccc ggtgtatagc	2460
tttctgccgg gcgtgtatag cgtgatttat ctgtatctga ccttttatct gaccaacgat	2520
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attaccattg cgtatattat ttgcattagc accaaacatt tttattggtt ttttagcaac	2640
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ctgtgcacct ttctgctgaa caaagaaatg tatctgaaac tgcgcagcga tgtgctgctg	2760
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gcgatggata ccaccagcta tcgcgaagcg gcgtgctgcc atctggcgaa agcgtgaa	2880
gatttttagca acagcggcag cgatgtgctg tatcagccgc cgcagaccag cattaccagc	2940
gcggtgctgc agagcggctt tcgcaaaatg gcgtttccga gcggcaaatg ggaaggctgc	3000
atggtgcagg tgacctgcgg caccaccacc ctgaacggcc tgtggctgga tgatgtggtg	3060
tattgccgcg gccatgtgat ttgcaccagc gaagatatgc tgaaccgaa ctatgaagat	3120
ctgctgattc gcaaaagcaa ccataacttt ctggtgcagg cgggcaacgt gcagctgcgc	3180

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gtgattggcc atagcatgca gaactgcgtg ctgaaactga aagtggtatc cgcgaacccg 3240
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tgctataacg gcagcccagc cggcgtgtat cagtgcgcga tgcgcccga ctttaccatt 3360
aaaggcagct ttctgaacgg cagctgcggc agcgtgggct ttaacattga ttatgattgc 3420
gtgagctttt gctatatgca tcatatggaa ctgccgaccg gcgtgcatgc gggcaccgat 3480
ctggaaggca acttttatgg ccggtttgtg gatcgccaga ccgcgcaggc ggcgggcacc 3540
gataccacca ttaccgtgaa cgtgctggcg tggctgtatg cggcgggtgat taacggcgat 3600
cgctggtttc tgaaccgctt taccaccacc ctgaacgatt ttaacctggt ggcgatgaaa 3660
tataactatg aaccgctgac ccaggatcat gtggatatc tgggcccgct gagcgcgcag 3720
accggcattg cggtgctgga tatgtgcgcg agcctgaaag aactgctgca gaacggcatg 3780
aacggccgca ccattctggg cagcgcgctg ctggaagatg aatttacctt gtttgatgtg 3840
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catcattggc tgctgctgac cattctgacc agcctgctgg tgctgggtgca gagcaccag 3960
tggagcctgt tttttttttt ttatgaaaac gcgtttctgc cgtttgcgat gggcattatt 4020
gcgatgagcg cgtttgcgat gatgtttgtg aaacataaac atgcgtttct gtgcctgttt 4080
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gcccctgctg aggaccagc cggcacagtc gcccacaga tccctcccaa ttggcacatt 4320
cccagcattc aggacgcagc aacccctac caccaccag caaccctaa taatatgggc 4380
ctgatcgtag gcgcagtcgg aggcctccctg ctggcagcac tggatcatctg tgggatcgtc 4440
tactggatga ggcggcggac acagaaggca ccaaagcgga ttccggtgcc acacattcgg 4500
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<210> SEQ ID NO 104

<211> LENGTH: 1513

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Covid19_gD_NSa (protein)

<400> SEQUENCE: 104

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Lys Leu Thr Met Met Gly Gly Ala Ala Ala Arg Leu Gly Ala Val Ile
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Leu Phe Val Val Ile Val Gly Leu His Gly Val Arg Gly Lys Tyr Ala
20          25          30
Leu Ala Asp Ala Ser Leu Lys Met Ala Asp Pro Asn Arg Phe Arg Gly
35          40          45
Lys Asp Leu Pro Val Leu Asp Gln Leu Thr Asp Pro Pro Gly Val Arg
50          55          60
Arg Val Tyr His Ile Gln Ala Gly Leu Pro Asp Pro Phe Gln Pro Pro
65          70          75          80
Ser Leu Pro Ile Thr Val Tyr Tyr Ala Val Leu Glu Arg Ala Cys Arg
85          90          95
Ser Val Leu Leu Asn Ala Pro Ser Glu Ala Pro Gln Ile Val Arg Gly

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100							105					110				
Ala	Ser	Glu	Asp	Val	Arg	Lys	Gln	Pro	Tyr	Asn	Leu	Thr	Ile	Ala	Trp	
		115					120					125				
Phe	Arg	Met	Gly	Gly	Asn	Cys	Ala	Ile	Pro	Ile	Thr	Val	Met	Glu	Tyr	
	130					135					140					
Thr	Glu	Cys	Ser	Tyr	Asn	Lys	Ser	Leu	Gly	Ala	Cys	Pro	Ile	Arg	Thr	
145					150					155					160	
Gln	Pro	Arg	Trp	Asn	Tyr	Tyr	Asp	Ser	Phe	Ser	Ala	Val	Ser	Glu	Asp	
				165					170					175		
Asn	Leu	Gly	Phe	Leu	Met	His	Ala	Pro	Ala	Phe	Glu	Thr	Ala	Gly	Thr	
			180				185						190			
Tyr	Leu	Arg	Leu	Val	Lys	Ile	Asn	Asp	Trp	Thr	Glu	Ile	Thr	Gln	Phe	
		195					200					205				
Ile	Leu	Glu	His	Arg	Ala	Lys	Gly	Ser	Cys	Lys	Tyr	Ala	Leu	Pro	Leu	
	210					215					220					
Arg	Ile	Pro	Pro	Ser	Ala	Cys	Leu	Ser	Pro	Gln	Ala	Tyr	Gln	Gln	Gly	
225					230					235					240	
Val	Thr	Val	Asp	Ser	Ile	Gly	Met	Leu	Pro	Arg	Phe	Ile	Pro	Glu	Asn	
				245					250					255		
Gln	Arg	Thr	Val	Ala	Val	Tyr	Ser	Leu	Lys	Ile	Ala	Gly	Trp	His	Gly	
			260					265					270			
Ser	Ser	Ala	Lys	Ser	Ala	Ser	Val	Tyr	Tyr	Ser	Gln	Leu	Met	Cys	Gln	
		275					280					285				
Pro	Ile	Leu	Leu	Leu	Asp	Gln	Ala	Leu	Val	Ser	Asp	Val	Gly	Asp	Ser	
	290					295					300					
Ala	Glu	Val	Ala	Val	Lys	Met	Phe	Asp	Ala	Tyr	Val	Asn	Thr	Phe	Ser	
305					310					315					320	
Ser	Thr	Phe	Asn	Val	Pro	Met	Glu	Lys	Leu	Lys	Thr	Leu	Val	Ala	Thr	
				325					330					335		
Ala	Glu	Ala	Glu	Leu	Ala	Lys	Asn	Val	Ser	Leu	Asp	Asn	Val	Leu	Ser	
			340				345						350			
Thr	Phe	Ile	Ser	Ala	Ala	Arg	Gln	Gly	Phe	Val	Asp	Ser	Asp	Val	Glu	
		355					360					365				
Thr	Lys	Asp	Val	Val	Glu	Cys	Leu	Lys	Leu	Ser	His	Gln	Ser	Asp	Ile	
	370					375					380					
Glu	Val	Thr	Gly	Asp	Ser	Cys	Asn	Asn	Tyr	Met	Leu	Thr	Tyr	Asn	Lys	
385					390					395					400	
Val	Glu	Asn	Met	Thr	Pro	Arg	Asp	Leu	Gly	Ala	Cys	Ile	Asp	Cys	Ser	
				405					410					415		
Ala	Arg	His	Ile	Asn	Ala	Gln	Val	Ala	Lys	Ser	His	Asn	Ile	Ala	Leu	
							425						430			
Ile	Trp	Asn	Val	Lys	Asp	Phe	Met	Ser	Leu	Ser	Glu	Gln	Leu	Arg	Lys	
		435					440					445				
Gln	Ile	Arg	Ser	Ala	Ala	Lys	Lys	Asn	Asn	Leu	Pro	Phe	Lys	Leu	Thr	
	450					455					460					
Cys	Ala	Thr	Thr	Arg	Gln	Val	Val	Asn	Val	Val	Thr	Thr	Lys	Ile	Ala	
465					470					475					480	
Leu	Lys	Gly	Gly	Lys	Ile	Val	Asn	Asn	Trp	Leu	Lys	Gln	Leu	Ile	Lys	
				485					490					495		
Val	Thr	Leu	Val	Phe	Leu	Phe	Val	Ala	Ala	Ile	Phe	Tyr	Leu	Ile	Thr	
			500					505					510			

Pro	Val	His	Val	Met	Ser	Lys	His	Thr	Asp	Phe	Ser	Ser	Glu	Ile	Ile
	515						520					525			
Gly	Tyr	Lys	Ala	Ile	Asp	Gly	Gly	Val	Thr	Arg	Asp	Ile	Ala	Ser	Thr
	530					535					540				
Asp	Thr	Cys	Phe	Ala	Asn	Lys	His	Ala	Asp	Phe	Asp	Thr	Trp	Phe	Ser
545					550					555					560
Gln	Arg	Gly	Gly	Ser	Tyr	Thr	Asn	Asp	Lys	Ala	Cys	Pro	Leu	Ile	Ala
				565					570					575	
Ala	Val	Ile	Thr	Arg	Glu	Val	Gly	Phe	Val	Val	Pro	Gly	Leu	Pro	Gly
				580				585					590		
Thr	Ile	Leu	Arg	Thr	Thr	Asn	Gly	Asp	Phe	Leu	His	Phe	Leu	Pro	Arg
							600					605			
Val	Phe	Ser	Ala	Val	Gly	Asn	Ile	Cys	Tyr	Thr	Pro	Ser	Lys	Leu	Ile
	610					615					620				
Glu	Tyr	Thr	Asp	Phe	Ala	Thr	Ser	Ala	Cys	Val	Leu	Ala	Ala	Glu	Cys
625					630					635					640
Thr	Ile	Phe	Lys	Asp	Ala	Ser	Gly	Lys	Pro	Val	Pro	Tyr	Cys	Tyr	Asp
				645					650					655	
Thr	Asn	Val	Leu	Glu	Gly	Ser	Val	Ala	Tyr	Glu	Ser	Leu	Arg	Pro	Asp
				660				665					670		
Thr	Arg	Tyr	Val	Leu	Met	Asp	Gly	Ser	Ile	Ile	Gln	Phe	Pro	Asn	Thr
							680					685			
Tyr	Leu	Glu	Gly	Ser	Val	Arg	Val	Val	Thr	Thr	Phe	Asp	Ser	Glu	Tyr
	690					695					700				
Cys	Arg	His	Gly	Thr	Cys	Glu	Arg	Ser	Glu	Ala	Gly	Val	Cys	Val	Ser
705					710					715					720
Thr	Ser	Gly	Arg	Trp	Val	Leu	Asn	Asn	Asp	Tyr	Tyr	Arg	Ser	Leu	Pro
				725					730					735	
Gly	Val	Phe	Cys	Gly	Val	Asp	Ala	Val	Asn	Leu	Leu	Thr	Asn	Met	Phe
				740				745					750		
Thr	Pro	Leu	Ile	Gln	Pro	Ile	Gly	Ala	Leu	Asp	Ile	Ser	Ala	Ser	Ile
		755					760					765			
Val	Ala	Gly	Gly	Ile	Val	Ala	Ile	Val	Val	Thr	Cys	Leu	Ala	Tyr	Tyr
	770					775					780				
Phe	Met	Arg	Phe	Arg	Arg	Ala	Phe	Gly	Glu	Tyr	Ser	His	Val	Val	Ala
785					790					795					800
Phe	Asn	Thr	Leu	Leu	Phe	Leu	Met	Ser	Phe	Thr	Val	Leu	Cys	Leu	Thr
				805					810					815	
Pro	Val	Tyr	Ser	Phe	Leu	Pro	Gly	Val	Tyr	Ser	Val	Ile	Tyr	Leu	Tyr
				820				825					830		
Leu	Thr	Phe	Tyr	Leu	Thr	Asn	Asp	Val	Ser	Phe	Leu	Ala	His	Ile	Gln
				835			840					845			
Trp	Met	Val	Met	Phe	Thr	Pro	Leu	Val	Pro	Phe	Trp	Ile	Thr	Ile	Ala
	850					855					860				
Tyr	Ile	Ile	Cys	Ile	Ser										

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Lys	Leu	Arg	Ser	Asp	Val	Leu	Leu	Pro	Leu	Thr	Gln	Tyr	Asn	Arg	Tyr
	915						920					925			
Leu	Ala	Leu	Tyr	Asn	Lys	Tyr	Lys	Tyr	Phe	Ser	Gly	Ala	Met	Asp	Thr
	930					935					940				
Thr	Ser	Tyr	Arg	Glu	Ala	Ala	Cys	Cys	His	Leu	Ala	Lys	Ala	Leu	Asn
945				950						955					960
Asp	Phe	Ser	Asn	Ser	Gly	Ser	Asp	Val	Leu	Tyr	Gln	Pro	Pro	Gln	Thr
			965					970						975	
Ser	Ile	Thr	Ser	Ala	Val	Leu	Gln	Ser	Gly	Phe	Arg	Lys	Met	Ala	Phe
			980					985					990		
Pro	Ser	Gly	Lys	Val	Glu	Gly	Cys	Met	Val	Gln	Val	Thr	Cys	Gly	Thr
		995					1000					1005			
Thr	Thr	Leu	Asn	Gly	Leu	Trp	Leu	Asp	Asp	Val	Val	Tyr	Cys	Pro	
1010						1015					1020				
Arg	His	Val	Ile	Cys	Thr	Ser	Glu	Asp	Met	Leu	Asn	Pro	Asn	Tyr	
1025						1030					1035				
Glu	Asp	Leu	Leu	Ile	Arg	Lys	Ser	Asn	His	Asn	Phe	Leu	Val	Gln	
1040						1045					1050				
Ala	Gly	Asn	Val	Gln	Leu	Arg	Val	Ile	Gly	His	Ser	Met	Gln	Asn	
1055						1060					1065				
Cys	Val	Leu	Lys	Leu	Lys	Val	Asp	Thr	Ala	Asn	Pro	Lys	Thr	Pro	
1070						1075					1080				
Lys	Tyr	Lys	Phe	Val	Arg	Ile	Gln	Pro	Gly	Gln	Thr	Phe	Ser	Val	
1085						1090					1095				
Leu	Ala	Cys	Tyr	Asn	Gly	Ser	Pro	Ser	Gly	Val	Tyr	Gln	Cys	Ala	
1100						1105					1110				
Met	Arg	Pro	Asn	Phe	Thr	Ile	Lys	Gly	Ser	Phe	Leu	Asn	Gly	Ser	
1115						1120					1125				
Cys	Gly	Ser	Val	Gly	Phe	Asn	Ile	Asp	Tyr	Asp	Cys	Val	Ser	Phe	
1130						1135					1140				
Cys	Tyr	Met	His	His	Met	Glu	Leu	Pro	Thr	Gly	Val	His	Ala	Gly	
1145						1150					1155				
Thr	Asp	Leu	Glu	Gly	Asn	Phe	Tyr	Gly	Pro	Phe	Val	Asp	Arg	Gln	
1160						1165					1170				
Thr	Ala	Gln	Ala	Ala	Gly	Thr	Asp	Thr	Thr	Ile	Thr	Val	Asn	Val	
1175						1180					1185				
Leu	Ala	Trp	Leu	Tyr	Ala	Ala	Val	Ile	Asn	Gly	Asp	Arg	Trp	Phe	
1190						1195					1200				
Leu	Asn	Arg	Phe	Thr	Thr	Thr	Leu	Asn	Asp	Phe	Asn	Leu	Val	Ala	
1205						1210					1215				
Met	Lys	Tyr	Asn	Tyr	Glu	Pro	Leu	Thr	Gln	Asp	His	Val	Asp	Ile	
1220						1225					1230				
Leu	Gly	Pro	Leu	Ser	Ala	Gln	Thr	Gly	Ile	Ala	Val	Leu	Asp	Met	
1235						1240					1245				
Cys	Ala	Ser	Leu	Lys	Glu	Leu	Leu	Gln	Asn	Gly	Met	Asn	Gly	Arg	
1250						1255					1260				
Thr	Ile	Leu	Gly	Ser	Ala	Leu	Leu	Glu	Asp	Glu	Phe	Thr	Pro	Phe	
1265						1270					1275				
Asp	Val	Val	Arg	Gln	Cys	Ser	Gly	Val	Thr	Phe	Gln	Ser	Ala	Val	
1280						1285					1290				
Lys	Arg	Thr	Ile	Lys	Gly	Thr	His	His	Trp	Leu	Leu	Leu	Thr	Ile	

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1295	1300	1305
Leu Thr Ser Leu Leu Val	Leu Val Gln Ser Thr Gln	Trp Ser Leu
1310	1315	1320
Phe Phe Phe Phe Tyr Glu Asn	Ala Phe Leu Pro Phe	Ala Met Gly
1325	1330	1335
Ile Ile Ala Met Ser Ala Phe	Ala Met Met Phe Val	Lys His Lys
1340	1345	1350
His Ala Phe Leu Cys Leu Phe	Leu Leu Pro Ser Leu	Ala Thr Val
1355	1360	1365
Ala Tyr Phe Asn Met Val Tyr	Met Pro Ala Ser Trp	Val Met Arg
1370	1375	1380
Ile Gly Ser Ser Arg Lys Ala	Pro Tyr Thr Ser Thr	Leu Leu Pro
1385	1390	1395
Pro Glu Leu Ser Glu Thr Pro	Asn Ala Thr Gln Pro	Glu Leu Ala
1400	1405	1410
Pro Glu Asp Pro Glu Asp Ser	Ala Leu Leu Glu Asp	Pro Val Gly
1415	1420	1425
Thr Val Ala Pro Gln Ile Pro	Pro Asn Trp His Ile	Pro Ser Ile
1430	1435	1440
Gln Asp Ala Ala Thr Pro Tyr	His Pro Pro Ala Thr	Pro Asn Asn
1445	1450	1455
Met Gly Leu Ile Ala Gly Ala	Val Gly Gly Ser Leu	Leu Ala Ala
1460	1465	1470
Leu Val Ile Cys Gly Ile Val	Tyr Trp Met Arg Arg	Arg Thr Gln
1475	1480	1485
Lys Ala Pro Lys Arg Ile Arg	Leu Pro His Ile Arg	Glu Asp Asp
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Gln Pro Ser Ser His Gln Pro	Leu Phe Tyr	
1505	1510	

<210> SEQ ID NO 105

<211> LENGTH: 4326

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Covid19_gD_Nsb (DNA) (5'-->3')

<400> SEQUENCE: 105

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gcagacccta acagattcag agggaaagac ctgccctgctc tggatcagct gactgatccc      180
ccaggcgtga gacgggtgta tcatatccaa gctggcctgc cagatccatt ccagccccc      240
tccctgccta tcaccgtcta ctacgcagtg ctggagagag cctgtaggtc cgtgctgctg      300
aatgtccta gcgaagcccc tcaaatgtg cggggcgcct ccgaagcgt gagaaagcag      360
ccatataatc tgaccatgcg ttggtttagg atgggcggca actgcgccat tcctatcacc      420
gtgatggagt atactgaatg cagctataac aagtccttgg gcgcctgccc aattagaacc      480
caacctaggt ggaactatta tgacagcttt tccgcagtca gcgaagacaa cctgggggtc      540
ctgatgcacg ccccgctttt tgaaaactgcc gggacatata tgccgctggg caaaatcaac      600
gattggacag aaatcaccca gttcatcctg gaacataggg ccaaaggagg ctgcaaatat      660

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gcccctgccac	tgaggatccc	cccttccgct	tgtctgtccc	ctcaagccta	ccagcaagga	720
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gctgtgtact	ccctgaagat	tgccggatgg	cacggatcca	gcggcgctgc	ggtgggtgat	840
agctattata	gcctgctgat	gccgattctg	accctgaccc	gcgcgctgac	cgcggaagc	900
catgtggata	ccgatctgac	caaaccgtat	attaaatggg	atctgctgaa	atatgatttt	960
accgaagaac	gcctgaaact	gtttgatcgc	tattttaaat	attgggatca	gacctatcat	1020
ccgaactcgc	tgaactgcct	ggatgatcgc	tgcattctgc	attgcgcgaa	ctttaacgtg	1080
ctgttttagca	ccgtgtttcc	gccgaccagc	tttggcccg	tggtgcgcaa	aatttttgtg	1140
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gcgaacgaat	gcgcgcaggt	gctgagcgaa	atggtgatgt	gcggcggcag	cctgtatgtg	2160
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accggccata	tgtcggatat	gtatagcgtg	atgctgacca	acgataaac	cagccgctat	2880
tgggaaccgg	aattttatga	agcgatgtat	accccgcata	ccgtgctgca	ggcgggtggc	2940

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gcgtgcgtgc tgtgcaacag ccagaccagc ctgcgtgcgc gcgcgtgcat tcgccgcccg 3000
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agcgtgaacc cgtatgtgtg caacgcgccg ggctgcgatg tgaccgatgt gaccagctg 3120
tatctgggcg gcatgagcta ttattgcaaa agccataaac cgccgattag ctttccgctg 3180
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accgatatta acgcgattgc gacctgcgat tggaccaacg cgggcgatta tattctggcg 3300
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gtcatctgtg ggatcgtcta ctggatgagg cggcggacac agaaggcacc aaagcggatt 4260
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<210> SEQ ID NO 106

<211> LENGTH: 1439

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Covid19_gD_Nsb (protein)

<400> SEQUENCE: 106

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Lys Leu Thr Met Met Gly Gly Ala Ala Ala Arg Leu Gly Ala Val Ile
1           5           10          15

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Leu Phe Val Val Ile Val Gly Leu His Gly Val Arg Gly Lys Tyr Ala
20          25          30

```

```

Leu Ala Asp Ala Ser Leu Lys Met Ala Asp Pro Asn Arg Phe Arg Gly
35          40          45

```

```

Lys Asp Leu Pro Val Leu Asp Gln Leu Thr Asp Pro Pro Gly Val Arg
50          55          60

```

```

Arg Val Tyr His Ile Gln Ala Gly Leu Pro Asp Pro Phe Gln Pro Pro
65          70          75          80

```

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Ser Leu Pro Ile Thr Val Tyr Tyr Ala Val Leu Glu Arg Ala Cys Arg
85          90          95

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Ser	Val	Leu	Leu	Asn	Ala	Pro	Ser	Glu	Ala	Pro	Gln	Ile	Val	Arg	Gly
			100					105					110		
Ala	Ser	Glu	Asp	Val	Arg	Lys	Gln	Pro	Tyr	Asn	Leu	Thr	Ile	Ala	Trp
		115					120					125			
Phe	Arg	Met	Gly	Gly	Asn	Cys	Ala	Ile	Pro	Ile	Thr	Val	Met	Glu	Tyr
	130					135					140				
Thr	Glu	Cys	Ser	Tyr	Asn	Lys	Ser	Leu	Gly	Ala	Cys	Pro	Ile	Arg	Thr
145					150					155					160
Gln	Pro	Arg	Trp	Asn	Tyr	Tyr	Asp	Ser	Phe	Ser	Ala	Val	Ser	Glu	Asp
				165					170					175	
Asn	Leu	Gly	Phe	Leu	Met	His	Ala	Pro	Ala	Phe	Glu	Thr	Ala	Gly	Thr
			180					185					190		
Tyr	Leu	Arg	Leu	Val	Lys	Ile	Asn	Asp	Trp	Thr	Glu	Ile	Thr	Gln	Phe
	195						200					205			
Ile	Leu	Glu	His	Arg	Ala	Lys	Gly	Ser	Cys	Lys	Tyr	Ala	Leu	Pro	Leu
	210					215					220				
Arg	Ile	Pro	Pro	Ser	Ala	Cys	Leu	Ser	Pro	Gln	Ala	Tyr	Gln	Gln	Gly
225					230					235					240
Val	Thr	Val	Asp	Ser	Ile	Gly	Met	Leu	Pro	Arg	Phe	Ile	Pro	Glu	Asn
				245					250					255	
Gln	Arg	Thr	Val	Ala	Val	Tyr	Ser	Leu	Lys	Ile	Ala	Gly	Trp	His	Gly
			260					265					270		
Ser	Ser	Gly	Val	Pro	Val	Val	Asp	Ser	Tyr	Tyr	Ser	Leu	Leu	Met	Pro
		275					280					285			
Ile	Leu	Thr	Leu	Thr	Arg	Ala	Leu	Thr	Ala	Glu	Ser	His	Val	Asp	Thr
	290					295					300				
Asp	Leu	Thr	Lys	Pro	Tyr	Ile	Lys	Trp	Asp	Leu	Leu	Lys	Tyr	Asp	Phe
305					310					315					320
Thr	Glu	Glu	Arg	Leu	Lys	Leu	Phe	Asp	Arg	Tyr	Phe	Lys	Tyr	Trp	Asp
				325					330					335	
Gln	Thr	Tyr	His	Pro	Asn	Cys	Val	Asn	Cys	Leu	Asp	Asp	Arg	Cys	Ile
			340					345					350		
Leu	His	Cys	Ala	Asn	Phe	Asn	Val	Leu	Phe	Ser	Thr	Val	Phe	Pro	Pro
		355					360					365			
Thr	Ser	Phe	Gly	Pro	Leu	Val	Arg	Lys	Ile	Phe	Val	Asp	Gly	Val	Pro
	370					375					380				
Phe	Val	Val	Ser	Thr	Gly	Tyr	His	Phe	Arg	Glu	Leu	Gly	Val	Val	His
385					390					395					400
Asn	Gln	Asp	Val	Asn	Leu	His	Ser	Ser	Arg	Leu	Ser	Phe	Lys	Glu	Leu
			405						410					415	
Leu	Val	Tyr	Ala	Ala	Asp	Pro	Ala	Met	His	Ala	Ala	Ser	Gly	Asn	Leu
			420					425					430		
Leu	Leu	Asp	Lys	Arg	Thr	Thr	Cys	Phe	Ser	Val	Ala	Ala	Leu	Thr	Asn
		435					440					445			
Asn	Val	Ala	Phe	Gln	Thr	Val	Lys	Pro	Gly	Asn	Phe	Asn	Lys	Asp	Phe
	450					455					460				
Tyr	Asp	Phe	Ala	Val	Ser	Lys	Gly	Phe	Phe	Lys	Glu	Gly	Ser	Ser	Val
465					470					475					480
Glu	Leu	Lys	His	Phe	Phe	Phe	Ala	Gln	Asp	Gly	Asn	Ala	Ala	Ile	Ser
			485						490					495	
Asp	Tyr	Asp	Tyr	Tyr	Arg	Tyr	Asn	Leu	Pro	Thr	Met	Cys	Asp	Ile	Arg

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500						505						510					
Gln	Leu	Leu	Phe	Val	Val	Glu	Val	Val	Asp	Lys	Tyr	Phe	Asp	Cys	Tyr		
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Asp	Gly	Gly	Cys	Ile	Asn	Ala	Asn	Gln	Val	Ile	Val	Asn	Asn	Leu	Asp		
530						535						540					
Lys	Ser	Ala	Gly	Phe	Pro	Phe	Asn	Lys	Trp	Gly	Lys	Ala	Arg	Leu	Tyr		
545						550						555					
Tyr	Asp	Ser	Met	Ser	Tyr	Glu	Asp	Gln	Asp	Ala	Leu	Phe	Ala	Tyr	Thr		
565						570						575					
Lys	Arg	Asn	Val	Ile	Pro	Thr	Ile	Thr	Gln	Met	Asn	Leu	Lys	Tyr	Ala		
580						585						590					
Ile	Ser	Ala	Lys	Asn	Arg	Ala	Arg	Thr	Val	Ala	Gly	Val	Ser	Ile	Cys		
595						600						605					
Ser	Thr	Met	Thr	Asn	Arg	Gln	Phe	His	Gln	Lys	Leu	Leu	Lys	Ser	Ile		
610						615						620					
Ala	Ala	Thr	Arg	Gly	Ala	Thr	Val	Val	Ile	Gly	Thr	Ser	Lys	Phe	Tyr		
625						630						635					
Gly	Gly	Trp	His	Asn	Met	Leu	Lys	Thr	Val	Tyr	Ser	Asp	Val	Glu	Asn		
645						650						655					
Pro	His	Leu	Met	Gly	Trp	Asp	Tyr	Pro	Lys	Cys	Asp	Arg	Ala	Met	Pro		
660						665						670					
Asn	Met	Leu	Arg	Ile	Met	Ala	Ser	Leu	Val	Leu	Ala	Arg	Lys	His	Thr		
675						680						685					
Thr	Cys	Cys	Ser	Leu	Ser	His	Arg	Phe	Tyr	Arg	Leu	Ala	Asn	Glu	Cys		
690						695						700					
Ala	Gln	Val	Leu	Ser	Glu	Met	Val	Met	Cys	Gly	Gly	Ser	Leu	Tyr	Val		
705						710						715					
Lys	Pro	Gly	Gly	Thr	Ser	Ser	Gly	Asp	Ala	Thr	Thr	Ala	Tyr	Ala	Asn		
725						730						735					
Ser	Val	Phe	Asn	Ile	Cys	Gln	Ala	Val	Thr	Ala	Asn	Val	Asn	Ala	Leu		
740						745						750					
Leu	Ser	Thr	Asp	Gly	Asn	Lys	Ile	Ala	Asp	Lys	Tyr	Val	Arg	Asn	Leu		
755						760						765					
Gln	His	Arg	Leu	Tyr	Glu	Cys	Leu	Tyr	Arg	Asn	Arg	Asp	Val	Asp	Thr		
770						775						780					
Asp	Phe	Val	Asn	Glu	Phe	Tyr	Ala	Tyr	Leu	Arg	Lys	His	Phe	Ser	Met		
785						790						795					
Met	Ile	Leu	Ser	Asp	Asp	Ala	Val	Val	Cys	Phe	Asn	Ser	Thr	Tyr	Ala		
805						810						815					
Ser	Gln	Gly	Leu	Val	Ala	Ser	Ile	Lys	Asn	Phe	Lys	Ser	Val	Leu	Tyr		
820						825						830					
Tyr	Gln	Asn	Asn	Val	Phe	Met	Ser	Glu	Ala	Lys	Cys	Trp	Thr	Glu	Thr		
835						840						845					
Asp	Leu	Thr	Lys	Gly	Pro	His	Glu	Phe	Cys	Ser	Gln	His	Thr	Met	Leu		
850						855						860					
Val	Lys	Gln	Gly	Asp	Asp	Tyr	Val	Tyr	Leu	Pro	Tyr	Pro	Asp	Pro	Ser		
865						870						875					
Arg	Ile	Leu	Gly	Ala	Gly	Cys	Phe	Val	Asp	Asp	Ile	Val	Lys	Thr	Asp		
885						890						895					
Gly	Thr	Leu	Met	Ile	Glu	Arg	Phe	Val	Ser	Leu	Ala	Ile	Asp	Ala	Tyr		
900						905						910					

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Pro	Leu	Thr	Lys	His	Pro	Asn	Gln	Glu	Tyr	Ala	Asp	Val	Phe	His	Leu
	915						920					925			
Tyr	Leu	Gln	Tyr	Ile	Arg	Lys	Leu	His	Asp	Glu	Leu	Thr	Gly	His	Met
	930					935					940				
Leu	Asp	Met	Tyr	Ser	Val	Met	Leu	Thr	Asn	Asp	Asn	Thr	Ser	Arg	Tyr
945					950					955					960
Trp	Glu	Pro	Glu	Phe	Tyr	Glu	Ala	Met	Tyr	Thr	Pro	His	Thr	Val	Leu
			965						970					975	
Gln	Ala	Val	Gly	Ala	Cys	Val	Leu	Cys	Asn	Ser	Gln	Thr	Ser	Leu	Arg
		980						985					990		
Cys	Gly	Ala	Cys	Ile	Arg	Arg	Pro	Phe	Leu	Cys	Cys	Lys	Cys	Cys	Tyr
	995						1000					1005			
Asp	His	Val	Ile	Ser	Thr	Ser	His	Lys	Leu	Val	Leu	Ser	Val	Asn	
	1010					1015					1020				
Pro	Tyr	Val	Cys	Asn	Ala	Pro	Gly	Cys	Asp	Val	Thr	Asp	Val	Thr	
	1025					1030					1035				
Gln	Leu	Tyr	Leu	Gly	Gly	Met	Ser	Tyr	Tyr	Cys	Lys	Ser	His	Lys	
	1040					1045					1050				
Pro	Pro	Ile	Ser	Phe	Pro	Leu	Cys	Ala	Asn	Gly	Gln	Val	Phe	Gly	
	1055					1060					1065				
Leu	Tyr	Lys	Asn	Thr	Cys	Val	Gly	Ser	Asp	Asn	Val	Thr	Asp	Phe	
	1070					1075					1080				
Asn	Ala	Ile	Ala	Thr	Cys	Asp	Trp	Thr	Asn	Ala	Gly	Asp	Tyr	Ile	
	1085					1090					1095				
Leu	Ala	Asn	Thr	Cys	Thr	Glu	Arg	Leu	Lys	Leu	Phe	Ala	Ala	Glu	
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Thr	Leu	Lys	Ala	Thr	Glu	Glu	Thr	Phe	Lys	Leu	Ser	Tyr	Gly	Ile	
	1115					1120					1125				
Ala	Thr	Val	Arg	Glu	Val	Leu	Ser	Asp	Arg	Glu	Leu	His	Leu	Ser	
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Trp	Glu	Val	Gly	Lys	Pro	Arg	Pro	Pro	Leu	Asn	Arg	Asn	Tyr	Val	
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Phe	Thr	Gly	Tyr	Arg	Val	Thr	Lys	Asn	Ser	Lys	Val	Gln	Ile	Gly	
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Glu	Tyr	Thr	Phe	Glu	Lys	Gly	Asp	Tyr	Gly	Asp	Ala	Val	Val	Tyr	
	1175					1180					1185				
Arg	Gly	Thr	Thr	Thr	Tyr	Lys	Leu	Asn	Val	Gly	Asp	Tyr	Phe	Val	
	1190					1195					1200				
Leu	Thr	Ser	His	Thr	Val	Met	Pro	Leu	Ser	Ala	Pro	Thr	Leu	Val	
	1205					1210					1215				
Pro	Gln	Glu	His	Tyr	Val	Arg	Ile	Thr	Gly	Leu	Tyr	Pro	Thr	Leu	
	1220					1225					1230				
Asn	Ile	Ser	Asp	Glu	Phe	Ser	Ser	Asn	Val	Ala	Asn	Tyr	Gln	Lys	
	1235					1240					1245				
Val	Gly	Met	Gln	Lys	Tyr	Ser	Thr	Leu	Gln	Gly	Pro	Pro	Gly	Thr	
	1250					1255					1260				
Gly	Lys	Ser	His	Phe	Ala	Ile	Gly	Leu	Ala	Leu	Tyr	Tyr	Pro	Ser	
	1265					1270					1275				
Ala	Arg	Ile	Val	Tyr	Thr	Ala	Cys	Ser	His	Ala	Ala	Val	Asp	Ala	
	1280					1285					1290				

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Leu Cys	Glu Lys	Ala Leu	Lys Tyr	Leu Pro	Ile Asp	Lys Cys	Ser
1295			1300		1305		
Arg Ile	Gly Ser	Ser Arg	Lys Ala	Pro Tyr	Thr Ser	Thr Leu	Leu
1310			1315		1320		
Pro Pro	Glu Leu	Ser Glu	Thr Pro	Asn Ala	Thr Gln	Pro Glu	Leu
1325			1330		1335		
Ala Pro	Glu Asp	Pro Glu	Asp Ser	Ala Leu	Leu Glu	Asp Pro	Val
1340			1345		1350		
Gly Thr	Val Ala	Pro Gln	Ile Pro	Pro Asn	Trp His	Ile Pro	Ser
1355			1360		1365		
Ile Gln	Asp Ala	Ala Thr	Pro Tyr	His Pro	Pro Ala	Thr Pro	Asn
1370			1375		1380		
Asn Met	Gly Leu	Ile Ala	Gly Ala	Val Gly	Gly Ser	Leu Leu	Ala
1385			1390		1395		
Ala Leu	Val Ile	Cys Gly	Ile Val	Tyr Trp	Met Arg	Arg Arg	Thr
1400			1405		1410		
Gln Lys	Ala Pro	Lys Arg	Ile Arg	Leu Pro	His Ile	Arg Glu	Asp
1415			1420		1425		
Asp Gln	Pro Ser	Ser His	Gln Pro	Leu Phe	Tyr		
1430			1435				

<210> SEQ ID NO 107
 <211> LENGTH: 27
 <212> TYPE: RNA
 <213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 107

agcgccagcg uguacuacag ccagcug

27

<210> SEQ ID NO 108
 <211> LENGTH: 24
 <212> TYPE: RNA
 <213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 108

guggccguga agauguucga cgcc

24

<210> SEQ ID NO 109
 <211> LENGTH: 33
 <212> TYPE: RNA
 <213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 109

uacgugaaca ccuucagcag caccuucac gug

33

<210> SEQ ID NO 110
 <211> LENGTH: 45
 <212> TYPE: RNA
 <213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 110

uucguggccg ccaucucua ccugaucacc cccgugcacg ugaug

45

<210> SEQ ID NO 111
 <211> LENGTH: 27
 <212> TYPE: RNA
 <213> ORGANISM: Sars-CoV-2

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<400> SEQUENCE: 111

uucagcgccg ugggcaacau cugcuac

27

<210> SEQ ID NO 112

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 112

gugcccuacu gcuacgacac caacgug

27

<210> SEQ ID NO 113

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 113

uucaugaggu ucaggagggc cuucggc

27

<210> SEQ ID NO 114

<211> LENGTH: 54

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 114

caguggaugg ugauguucac cccccuggug ccuucugga ucaccaucgc cuac

54

<210> SEQ ID NO 115

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 115

aggguggugu ucaacggcgu gagcuuc

27

<210> SEQ ID NO 116

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 116

gcccugugca ccuuccugcu gaacaag

27

<210> SEQ ID NO 117

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 117

uaccugaagc ugaggagcga cgugcug

27

<210> SEQ ID NO 118

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 118

accaguaca acagguaccu ggcccug

27

<210> SEQ ID NO 119

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<211> LENGTH: 26
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 119
cccccaagua caaguucgug aggauc 26

<210> SEQ ID NO 120
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 120
gugagcuucu gcuacaugca ccacaug 27

<210> SEQ ID NO 121
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 121
ggcaccgacc uggaggggcaa cuucuaac 27

<210> SEQ ID NO 122
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 122
cuggccuggc uguacgccgc cgugauc 27

<210> SEQ ID NO 123
<211> LENGTH: 69
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 123
cugcccuucg ccaugggcgau caucgccaug agcgccuucg ccaugauguu cgugaagcac 60
aagcacgcc 69

<210> SEQ ID NO 124
<211> LENGTH: 69
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 124
cugcccagcc uggccaccgu ggccuacuuc aacauggugu acaugcccgc cagcugggug 60
augaggauac 69

<210> SEQ ID NO 125
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 125
gugguggaca gcuacuacag ccugcug 27

<210> SEQ ID NO 126
<211> LENGTH: 33
<212> TYPE: RNA

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<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 126

aagaucuucg uggacggcgu gccuucgug gug 33

<210> SEQ ID NO 127

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 127

cugguguacg ccgccgaccc cgccaug 27

<210> SEQ ID NO 128

<211> LENGTH: 33

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 128

agcguggccg ccugaccaa caacguggcc uuc 33

<210> SEQ ID NO 129

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 129

aucagcgacu acgacuacua cagguac 27

<210> SEQ ID NO 130

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 130

uucguggugg agguggugga caaguac 27

<210> SEQ ID NO 131

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 131

aggcuguacu acgacagcau gagcuac 27

<210> SEQ ID NO 132

<211> LENGTH: 45

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 132

uucgccuaca ccaagaggaa cgugaucucc accaucaccc agaug 45

<210> SEQ ID NO 133

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 133

aggcaguucc accagaagcu gcugaag 27

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<210> SEQ ID NO 134
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 134
agggcccaugc ccaacaugcu gaggauc 27

<210> SEQ ID NO 135
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 135
agccugagcc acagguucua caggcug 27

<210> SEQ ID NO 136
<211> LENGTH: 57
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 136
gacaccgacu ucgugaacga guucuacgcc uaccugagga agcacuucag caugaug 57

<210> SEQ ID NO 137
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 137
cugcccuacc ccgaccccag caggau 27

<210> SEQ ID NO 138
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 138
cugaugaucg agagguucgu gagccug 27

<210> SEQ ID NO 139
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 139
caccuguacc ugcaguacau caggaag 27

<210> SEQ ID NO 140
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 140
cugaccaacg acaacaccag cagguac 27

<210> SEQ ID NO 141
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

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<400> SEQUENCE: 141

gugaccgacg ugacccagcu guaccug

27

<210> SEQ ID NO 142

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 142

augagcuacu acugcaagag ccacaag

27

<210> SEQ ID NO 143

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 143

uucccccugu gcgccaacgg ccaggug

27

<210> SEQ ID NO 144

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 144

aagcuguucg ccgccgagac ccugaag

27

<210> SEQ ID NO 145

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 145

gagaccuua agcugagcua cggauc

27

<210> SEQ ID NO 146

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 146

aaggugcaga ucggcgagua caccuuc

27

<210> SEQ ID NO 147

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 147

gugguguaca ggggcaccac caccuac

27

<210> SEQ ID NO 148

<211> LENGTH: 27

<212> TYPE: RNA

<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 148

augccccuga gcgccccac ccuggug

27

<210> SEQ ID NO 149

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<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 149
gugggcaugc agaaguacag cacccug                27

<210> SEQ ID NO 150
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 150
uucgccaucg gccuggcccu guacuac                27

<210> SEQ ID NO 151
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 151
uacaccgccu gcagccacgc cgccgug                27

<210> SEQ ID NO 152
<211> LENGTH: 27
<212> TYPE: RNA
<213> ORGANISM: Sars-CoV-2

<400> SEQUENCE: 152
cugcccaucg acaagugcag caggau                27

<210> SEQ ID NO 153
<211> LENGTH: 12
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Spacer RNA Sequence

<400> SEQUENCE: 153
ggcggcggca gc                                12

<210> SEQ ID NO 154
<211> LENGTH: 1974
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Polyepitope RNA sequence

<400> SEQUENCE: 154
agcgccagcg uguacuacag ccagcugggc ggcggcagcg uggccgugaa gauguucgac    60
gccggcggcg gcagcuacgu gaacaccuuc agcagcaccu ucaacguggg cggcggcagc    120
uucguggccg ccaucuucua ccugaucacc cccugcacg ugauggggcg cggcagcuuc    180
agcgccgugg gcaacaucug cuacggcggc ggcagcgugc ccuacugcua cgacaccaac    240
guggggcgcg gcagcuucau gagguucagg agggccuucg gcgggggcg cagccagugg    300
auggugaugu ucaccccccu ggugcccuuc uggauaccca ucgccuacgg cgggcgagc    360
aggguggugu ucaacggcgu gagcuucggc ggcggcagcg ccugugcac cuuccugcug    420
aacaaggcg cgggcagcua ccugaagcug aggagcgagc ugcuggggcg cggcagcacc    480

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caguacaaca gguaccuggc ccugggcggc ggcagcacc ccaaguacaa guucgugagg	540
aucggcggcg gcagcgugag cuucugcuac augcaccaca uggcggcgcg cagcggcacc	600
gaccuggagg gcaacuucua cggcggcgcg agccuggccu ggcuguacgc cgcgugauc	660
ggcgcggcga gccugcccuu cgccaugggc aucaucgcca ugagcgccuu cgccaugaug	720
uucgugaagc acaagcacgc cggcggcgcg agccugccca gccuggccac cguggccuac	780
uucaacaugg uguacaugcc cgcagcugg gugaugagga ucggcggcg cagcguggug	840
gacagcuacu acagccugcu gggcggcgcg agcaagaucu ucguggacgg cgugcccuuc	900
guggugggcg cgcgcagccu gguguacgcc gccgaccccg ccaugggcg cggcagcagc	960
guggcgcccc ugaccaacaa cguggccuuc ggcggcgga gcaucagcga cuacgacuac	1020
uacagguacg cgcggcgag cuucguggug gagguggugg acaaguacgg cgcggcgagc	1080
aggcuguacu acgacagcau gagcuacggc ggcggcagcu ucgccuacac caagaggaa	1140
gugaucccca ccaucaccca gaugggcgcg ggcagcaggc aguuccacca gaagcugcug	1200
aaggcggcg gcagcaggc caugcccaac augcugagga ucggcggcg cagcagccug	1260
agccacaggu ucuacaggcu gggcggcgcg agcgacaccg acuucguga cgaguucua	1320
gccuaccuga ggaagcacuu cagcaugaug ggcggcgga gccugccua ccccgacccc	1380
agcaggaucc gcggcgagc ccugaugauc gagagguucg ugagccuggg cgcggcgagc	1440
caccuguacc ugcaguacau caggaagggc ggcggcagcc ugaccaacga caacaccagc	1500
agguacggcg cgcgcagcgu gaccgacgug acccagcugu accugggcg cggcagcaug	1560
agcuacuacu gcaagagcca caaggcggc ggcagcuucc ccugugcgc caacggccag	1620
guggcggcg gcagcaagcu guucgcccgc gagaccuga agggcggcg cagcgagacc	1680
uucaagcuga gcuacggcau cggcggcgcg agcaaggugc agaucggcga guacaccuuc	1740
ggcgcggcga gcguggugua caggggcacc accaccuacg gcggcgagc caugcccuug	1800
agcgcaccca ccugggugg cgcgcgcagc gugggcaugc agaaguacag caccugggc	1860
ggcggcagcu ucgccaucgg ccuggcccuu uacuacggcg gcggcagcua caccgccugc	1920
agccacggcg ccuggggcg cgcagccug cccaucgaca agugcagcag gauc	1974

<210> SEQ ID NO 155

<211> LENGTH: 819

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: RNA Sequence of the glycoprotein D (gD)
n-terminal fragment

<400> SEQUENCE: 155

aagcuuacca ugaugggagg ggcugcugcc agacugggag cugugauucu guucgugguc	60
auuguggggc ugcauggagu gcgggggaaa uacgcucugg ccgaugcaag ccugaaaug	120
gcagaccua acagaauacg agggaaagac cugcccugc uggaucagcu gacugauccc	180
ccaggcguga gacgggugua ucauauccaa gcuggccugc cagauccauu ccagccccc	240
ucccugccua ucaccgucua cuacgcagug cuggagagag ccuguagguc cgugcugcug	300
aauguccua gcgaagcccc ucaaaauug cggggcgccu ccgaagacgu gagaaagcag	360
ccauauaauc ugaccaucgc uugguuuagg augggcggca acugcgccau uccuauacc	420
gugauggagu auacugaaug cagcuauaac aaguccugg gcgcugccc aaauagaacc	480

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caaccuaggu ggaacuauua ugacagcuuu uccgcaguca gcgaagacaa ccuggggguuc	540
cugaugcacg cccccgcuuu ugaaacugcc gggacauauc ugcggcuggu caaaaauaac	600
gauuggacag aaauccacca guucauccug gaacauaggg ccaaaggag cugcaaaau	660
gcccugccac ugaggauccc ccuuccgcu ugucugucc cucaagccua ccagcaagga	720
gugacagugg acagcaucgg caugcugccc agauucauuc ccgagaacca gagaacugug	780
gcuguguacu ccugaagau ugccggaugg cacggaucc	819

<210> SEQ ID NO 156
 <211> LENGTH: 396
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: RNA Sequence of the gD C-terminal fragment

<400> SEQUENCE: 156

ggauccucua gaaaggcccc uuacacuucc acacugcugc cccugagcu gagcgaaaca	60
cccaaugcca cccagccoga gcuggcuccc gaggaccccg aggcagcgc ccugcuggag	120
gaccagucg gcacagucgc cccacagauc ccucccauu ggcacauucc cagcauucag	180
gacgcagcaa ccccuacca cccaccagca accccuaaua auaugggccu gaucgcaggc	240
gcagucggag gcucccugcu ggcagcacug gucaucugug ggaucgucua cuggaugagg	300
cggcggacac agaaggcacc aaagcggauu cggcugccac acauucggga agacgaucag	360
ccuuccuccc accagccacu guuuuacuaa cucgag	396

<210> SEQ ID NO 157
 <211> LENGTH: 3189
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Covid19_gD_POLIIEP (RNA) (5'-->3')

<400> SEQUENCE: 157

aagcuuacca ugaugggagg ggcugcugcc agacugggag cugugauucu guucgugguc	60
auuguggggc ugcauggagu gcgggggaaa uacgcucugy ccgaugcaag ccugaaaug	120
gcagaccua acagauucag agggaaagac cugcccugc uggauagcu gacugauccc	180
ccaggcguga gacgggugua ucauauccaa gcuggccugc cagauccau ccagccccc	240
ucccugccua ucaccgucua cuacgcagug cuggagagag ccuguagguc cgugcugcug	300
aaugcuccua gcgaagcccc ucaaaugug cggggcgccu ccgaagacgu gagaaagcag	360
ccauauaauc ugaccaucgc uugguuuagg augggcgga acugcgccau uccuauacc	420
gugauggagu auacugaau cagcuauaac aaguccugy gcgcccugccc aaauagaacc	480
caaccuaggu ggaacuauua ugacagcuuu uccgcaguca gcgaagacaa ccuggggguuc	540
cugaugcacg cccccgcuuu ugaaacugcc gggacauauc ugcggcuggu caaaaauaac	600
gauuggacag aaauccacca guucauccug gaacauaggg ccaaaggag cugcaaaau	660
gcccugccac ugaggauccc ccuuccgcu ugucugucc cucaagccua ccagcaagga	720
gugacagugg acagcaucgg caugcugccc agauucauuc ccgagaacca gagaacugug	780
gcuguguacu ccugaagau ugccggaugg cacggaucca gcgccagcgu guacuacagc	840
cagcugggcy gcggcagcgu ggcgugaag auguucgacg ccggcgggcg cagcuacug	900

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aacaccuuca	gcagcaccuu	caacgugggc	ggcggcagcu	ucguggccgc	caucuucua	960
cugaucacc	ccgugcacgu	gauggggcgc	ggcagcuuca	gcgcgugggg	caacaucug	1020
uacggcggcg	gcagcgugcc	cuacugcuac	gacaccaacg	ugggcggcgg	cagcuucaug	1080
agguucagga	gggccuucgg	cggcggcggc	agccagugga	uggugauguu	cacccccug	1140
gugcccuuc	ggauccacau	cgccuacggc	ggcggcagca	gggugguguu	caacggcgug	1200
agcuucggcg	gcggcagcgc	ccugugcacc	uuccugcuga	acaagggcgg	cggcagcuac	1260
cugaagcuga	ggagcgacgu	gcuggggcgc	ggcagcacc	aguacaacag	guaccuggcc	1320
cuggggcggcg	gcagcacc	caaguacaag	uucgugagga	ucggcggcgg	cagcgugagc	1380
uucugcuaca	ugcaccacau	gggcggcggc	agcggcaccg	accuggaggg	caacuucua	1440
ggcggcggca	gccuggccug	gcuguacgcc	gccgugaucg	gcggcggcag	ccugcccuuc	1500
gccaugggca	ucaucgccau	gagcgccuuc	gccaugaugu	ucgugaagca	caagcacgcc	1560
ggcggcggca	gccugcccag	ccuggccacc	guggccuacu	ucaacauggu	guacaugccc	1620
gccagcuggg	ugaugaggau	cggcggcggc	agcugggugg	acagcuacua	cagccugcug	1680
ggcggcggca	gcaagaucuu	cguggacggc	gugcccuucg	ugguggggcg	cggcagccug	1740
guguacggcg	ccgaccccg	cauggggcgc	ggcagcagcg	uggccgcccu	gaccaacaac	1800
guggccuucg	gcggcggcag	caucagcgac	uacgacuacu	acagguacgg	cggcggcagc	1860
uucguggugg	agguggugga	caaguacggc	ggcggcagca	ggcuguacua	cgacagcaug	1920
agcuacggcg	gcggcagcuu	cgccuacacc	aagaggaacg	ugaucgccac	caucaccag	1980
auggggcggcg	gcagcaggca	guuccaccag	aagcugcuga	agggcggcgg	cagcagggcc	2040
augccaaca	ugcugaggau	cggcggcggc	agcagccuga	gccacagguu	cuacaggcug	2100
ggcggcggca	gcgacaccga	cuucgugaac	gaguucuaag	ccuaccugag	gaagcacuuc	2160
agcaugaugg	gcggcggcag	ccugcccuac	cccgaaccca	gcaggauccg	cggcggcagc	2220
cugaugaucg	agagguucgu	gagccugggc	ggcggcagcc	accuguaacu	gcaguacauc	2280
aggaaggggcg	gcggcagccu	gaccaaccgac	aacaccagca	gguacggcgg	cggcagcgug	2340
accgacguga	cccagcugua	ccuggggcgc	ggcagcauga	gcuacuacug	caagagccac	2400
aaggggcggcg	gcagcuuccc	ccugugcgcc	aacggccagg	ugggcggcgg	cagcaagcug	2460
uucgcccggcg	agacccugaa	gggcggcggc	agcgagaccu	ucaagcugag	cuacggcauc	2520
ggcggcggca	gcaaggugca	gaucggcgag	uacaccuucg	gcggcggcag	cugggugua	2580
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<210> SEQ ID NO 159

<211> LENGTH: 4326

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Covid19_gD_NSb (RNA) (5'-->3')

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<212> TYPE: PRT
<213> ORGANISM: Sars-CoV-2

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<400> SEQUENCE: 160

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1. A chimeric nucleotide sequence comprising the sequences encoding, in phase, a plurality of epitopes of a coronavirus non-structural protein.

2. The chimeric nucleotide sequence according to claim 1 comprising the sequences encoding multiple epitopes of replicase 1ab, aligned in phase.

3. The chimeric nucleotide sequence according to claim 1, wherein the nucleotide sequence encoding the polypeptide comprises a phased combination of the nucleotide sequences selected from SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 73, 75, 77, 79, 81, 83, 85, 87, 89, 91.

4. The chimeric nucleotide sequence according to claim 1, wherein the nucleotide sequence encoding the polypeptide is defined by SEQ ID NO: 95.

5. The chimeric nucleotide sequence according to claim 1, further comprising a nucleotide sequence that encodes a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

6. The chimeric nucleotide sequence according to claim 1, wherein the nucleotide sequence encoding the modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1) is defined by SEQ ID NO: 97 and/or SEQ ID NO: 99.

7. The chimeric nucleotide sequence according to claim 1, further comprising a nucleotide sequence encoding a first spacer polypeptide.

8. The chimeric nucleotide sequence according to claim 1, further comprising:

- a) a nucleotide sequence that encodes a first peptide comprising epitopes found in the amino acid sequence of the replicase polypeptide 1ab (PR1ab);
- b) a nucleotide sequence encoding a first spacer polypeptide; and

c) a nucleotide sequence that encodes a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

9. The chimeric nucleotide sequence according to claim 1, wherein the chimeric nucleotide sequence is defined by SEQ ID NO: 101.

10. A mammalian expression vector comprising a chimeric nucleotide sequence encoding the polypeptide, as defined in claim 1; and

one or more expression promoters functionally linked to said chimeric nucleotide sequence.

11. An RNA vaccine comprising an RNA sequence that encodes a plurality of epitopes of a coronavirus non-structural protein.

12. The RNA vaccine according to claim 11, further comprising the sequences encoding multiple epitopes of replicase 1ab, aligned in phase.

13. The RNA vaccine according to claim 11 comprising a combination of the RNA sequences selected from SEQ ID NO: 113 to SEQ ID NO: 158.

14. A chimeric fusion protein comprising a plurality of epitopes of a coronavirus non-structural protein.

15. The chimeric fusion protein according to claim 14, comprising a combination of the peptide sequences of replicase 1ab selected from SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92.

16. The chimeric fusion protein according to claim 14 comprising SEQ ID NO: 96.

17. The chimeric fusion protein according to claim 14, further comprising a polypeptide sequence of a modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

18. The chimeric fusion protein according to claim 17, wherein the polypeptide sequence of the modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1) is defined by SEQ ID NO: 98 and/or SEQ ID NO: 100.

19. The chimeric fusion protein according to claim **14**, further comprising a sequence of a first spacer polypeptide.

20. The chimeric fusion protein according to claim **14**, comprising:

- a) a first peptide consisting of epitopes found in the amino acid sequence of the replicase polyprotein 1ab (PR1ab);
- b) a first spacer; and
- c) modified form of the glycoprotein D (gD) of herpes simplex type 1 (HSV-1).

21. The chimeric fusion protein according to claim **14**, comprising the amino acid sequence defined by SEQ ID NO: 108.

22. A method for the preparation of a vaccine for coronavirus, SARS-COV-2 and/or related viruses comprising a chimeric nucleotide sequence as defined in claim **1** or a chimeric fusion protein as defined in claim **14**.

23. A method of preparation of a vaccine for coronavirus, SARS-COV-2, and/or related viruses, wherein the method comprises preparation of a chimeric nucleotide sequence as defined in claim **1** or a chimeric fusion protein as defined in claim **14**.

24. A method of inducing cell-type immune response comprising administering an effective dosage of a chimeric nucleotide sequence as defined in claim **1** or a chimeric fusion protein as defined in claim **14**.

25. A method of treatment of viral diseases comprising administering an effective dosage of a chimeric nucleotide sequence as defined in claim **1** or a chimeric fusion protein as defined in claim **14**.

* * * * *